STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – I

Simple Stresses and Strains

Handwritten Study Notes

Syllabus topics covered in this unit :

Elasticity and plasticity — Types of stresses and strains — Hooke's law — Factor of safety, Poisson's ratio — Relationship between elastic constants — Bars of varying section — Stresses in composite bars

Contents of these notes :

→ 1.1 Introduction – Strength of Materials
→ 1.2 Stress–Strain Curve (Mild Steel)
→ 1.3 Poisson's Ratio and Elastic Constants
→ 1.4 Bars of Varying Section
→ 1.5 Composite (Compound) Bars
→ 1.6 Shear, Bearing and Punching Stresses
→ 1.7 Statically Indeterminate Bars and Temperature
→ 1.8 More Solved Problems
→ 1.9 Important Short Questions (Exam Oriented)

1.1 Introduction – Strength of Materials

Strength of Materials (Mechanics of Solids) studies how a body responds to external loads – the **internal resistance** (stress) and the **deformation** (strain) it develops.

- Rigid body mechanics: body is assumed non-deformable (statics/dynamics).
- Deformable body mechanics: loads cause change of size/shape – our subject.
- Aim of design : stresses < permissible limit, deformations < allowable limit.

Elasticity and Plasticity

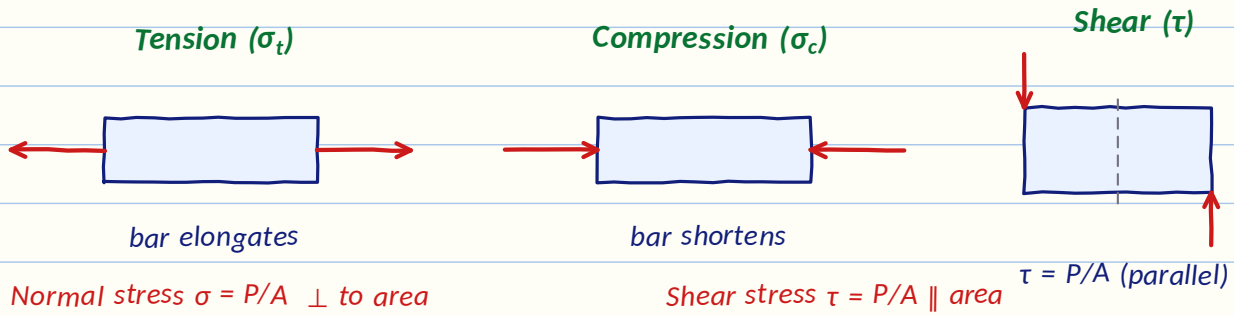
- **Elasticity** : property by which a body regains its original size and shape on removal of the load. Perfectly elastic – quartz, steel (within limit).
- **Plasticity** : body does not regain its shape – permanent (residual) deformation remains. e.g. lead, clay, steel beyond the yield point.
- Elastic limit : max stress up to which the body is perfectly elastic.
- Ductile : large plastic strain before failure (mild steel, copper). Brittle : fails with little strain (cast iron, glass, concrete).
- Malleability – can be hammered into sheets ; Toughness – energy absorbed up to fracture.

Stress

Internal resistance offered per unit area to the external load.

$$\sigma = \frac{P}{A} \qquad (N/mm^2 = MPa)$$

- $1\ N/mm^2 = 1\ MPa = 10^6\ N/m^2\ (Pa)$; $1\ GPa = 10^3\ MPa = 10^9\ Pa$.



Normal stress $\sigma = P/A$ $\perp$ to area

Shear stress $\tau = P/A$ $\parallel$ area

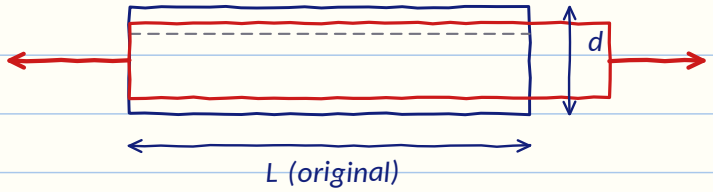
Types of Stresses

Type	Cause	Formula
Tensile	Pulling force	$\sigma_t = P/A$
Compressive	Pushing force	$\sigma_c = P/A$
Shear	Forces ∥ to area	$\tau = P/A$
Bearing	Contact pressure	$\sigma_b = P/(d \cdot t)$
Thermal	Restrained expansion	$\sigma = \alpha T E$
Volumetric (bulk)	Hydrostatic p	$\sigma = p$

Strain

Ratio of change in dimension to original dimension – dimensionless.

- Linear (longitudinal) strain $\varepsilon = \delta L/L$ (δL = change in length)
- Lateral strain = $\delta d/d$ – opposite in sign to longitudinal strain.
- Shear strain φ (γ) = angular distortion in radians = $\tan \varphi \approx \varphi = \delta/h$
- Volumetric strain $\varepsilon_v = \delta V/V = \varepsilon_x + \varepsilon_y + \varepsilon_z$



Longitudinal strain $\varepsilon = \delta L/L$ (+)

Lateral strain = $-\delta d/d$ (–)

Poisson's ratio μ = – lateral/longitudinal

μ = 0.25 – 0.33 for metals (max 0.5)

Hooke's Law

Within the elastic (proportional) limit, stress is directly proportional to strain.

$$\sigma \propto \varepsilon \Rightarrow \sigma = E\,\varepsilon \Rightarrow E = \frac{\sigma}{\varepsilon}$$

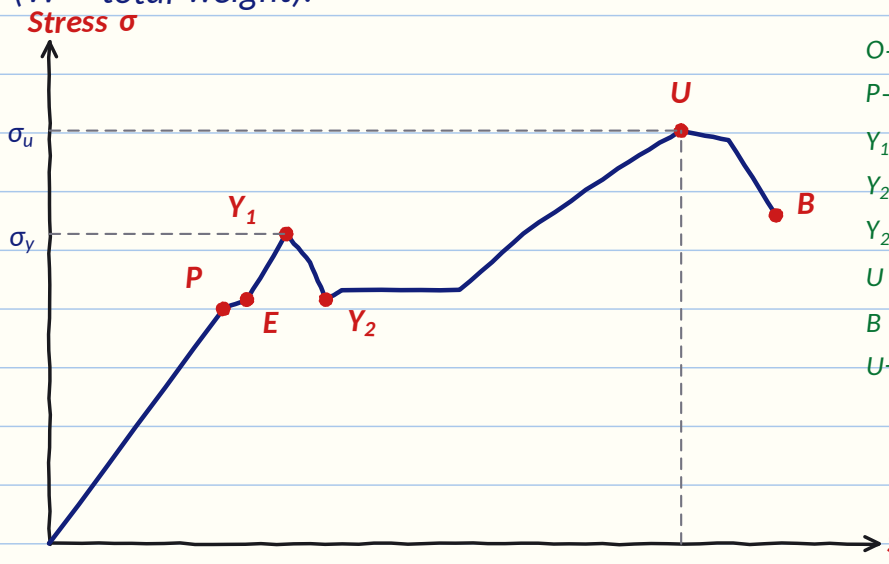
- E = Young's modulus (modulus of elasticity) – tensile/compressive.
- G (C, N) = Modulus of rigidity = τ/φ (shear).
- K = Bulk modulus = $p/(\delta V/V)$ (volumetric).

Material	E (GPa)	G (GPa)	μ
Mild steel	200–210	80	0.25–0.30
Cast iron	100–120	40	0.23–0.27
Copper	100–110	40	0.32–0.35
Aluminium	70	26	0.33

Deformation of a Bar under Axial Load

$$\delta L = \frac{P\,L}{A\,E}$$

- E A is the "axial rigidity" – larger E A means smaller elongation.
- Bar under its own weight (length L, weight density w): $\delta L = (w\,L^2)/2E = W\,L/2\,A\,E$ (W = total weight).



O–P : proportional limit
P–E : elastic limit
Y_1 : upper yield pt
Y_2 : lower yield pt
Y_2–: plastic flow
U : ultimate stress
B : breaking (fracture)
U–B : necking

1.2 Stress–Strain Curve (Mild Steel)

- Obtained by Universal Testing Machine (UTM) on a standard specimen.
- O–P : straight line – Hooke's law valid ; slope = E.
- P–E : slightly beyond P – still elastic but not proportional.
- Y_1 , Y_2 : yield – strain increases with almost no increase in load. Material starts to flow plastically. Upper yield Y_1 is sensitive ; use lower yield Y_2.
- $Y_2 \rightarrow U$: strain hardening – stress rises again up to ultimate stress.
- $U \rightarrow B$: necking – area reduces locally, load drops, specimen breaks at B.

Important results from the test

- Percentage elongation = $(L_f - L_0)/(L_0) \times 100$ (ductility measure)
- Percentage reduction in area = $(A_0 - A_f)/(A_0) \times 100$
- True stress = P / (instantaneous area) > nominal stress after yield.
- Proof stress : for materials with no definite yield point (aluminium, copper) take stress for 0.2 % permanent strain (draw a line ∥ to OP from 0.2 % strain).
- Cast iron (brittle): no yield point, fails suddenly, strong in compression.

Factor of Safety (F.O.S.)

$$F.O.S. = \frac{\text{Ultimate stress}}{\text{Working (permissible) stress}}$$

- For ductile materials, base on yield stress : F.O.S. = $\sigma_y / \sigma_{working}$.
- Typical values: steel 1.85–3 ; cast iron 4–6 ; timber 7–10 ; concrete 3–4.
- Margin of safety = F.O.S. – 1.
- Why? uncertainty in loads, material defects, stress concentration, workmanship.

Example 1

A steel rod 20 mm dia., 2 m long carries a tensile load 60 kN. $E = 2\times10^5$ N/mm^2. Find stress, strain and elongation.

$A = \pi/4 \times 20^2 = 314.16$ mm^2

$\sigma = P/A = 60000 / 314.16 =$ **190.99 N/mm^2**

$\varepsilon = \sigma/E = 190.99 / (2\times10^5) =$ **9.55×10^{-4}**

$\delta L = \varepsilon L = 9.55\times10^{-4} \times 2000 =$ **1.91 mm**

Example 2 (Factor of safety)

A bar of 25 mm dia. fails at 180 kN. If the working load is 45 kN, F.O.S. = ?

Ultimate load / working load = 180/45 = **4**

1.3 Poisson's Ratio and Elastic Constants

Poisson's ratio μ (or 1/m) = Lateral strain/Longitudinal strain (within elastic limit).

- Axial pull ⇒ length ↑, lateral dimensions ↓ (and vice versa). Value 0.25 to 0.33 for most metals; theoretical max 0.5.

Three-dimensional stress system

Strain in x-direction when σ_x, σ_y, σ_z (tension +) act:

$$\varepsilon_x = \frac{\sigma_x}{E} - \mu\frac{\sigma_y}{E} - \mu\frac{\sigma_z}{E}$$

- Similarly for ε_y and ε_z.
- Volumetric strain $\varepsilon_v = \varepsilon_x+\varepsilon_y+\varepsilon_z = (\sigma_x+\sigma_y+\sigma_z)/E(1-2\mu)$
- For a bar loaded in x-direction only : $\varepsilon_v = \varepsilon(1-2\mu)$; $\delta V = \varepsilon(1-2\mu)V$.
- Since $\varepsilon_v \geq 0 \Rightarrow \mu \leq 0.5$.

Relationship between E, G, K and μ

$$E = 2G(1+\mu)$$

$$E = 3K(1-2\mu)$$

$$E = \frac{9KG}{3K+G} \quad ; \quad \mu = \frac{3K-2G}{2(3K+G)}$$

Derivation of E = 2G(1+μ) (pure shear)

- Take a square element ABCD under shear stress τ on all four faces.
- Shear produces a diagonal tension τ and diagonal compression τ at 45° (principal stresses +τ and −τ).
- Strain of diagonal BD = $\tau/E + \mu\,\tau/E = \tau/E(1+\mu)$.
- From geometry, diagonal strain = ½ (shear strain φ) = $\tau/2G$.
- Equate: $\tau/2G = \tau/E(1+\mu) \Rightarrow$ **$E = 2G(1+\mu)$**

Derivation of E = 3K(1−2μ) (cube under p)

- Cube subjected to uniform pressure p on all faces: $\sigma_x=\sigma_y=\sigma_z = p$.
- Linear strain $\varepsilon = p/E(1-2\mu)$, volumetric strain $\varepsilon_v = 3\varepsilon = 3p/E(1-2\mu)$.
- By definition $K = p/\varepsilon_v \Rightarrow$ **$E = 3K(1-2\mu)$**
- Eliminating μ from the two results gives $E = 9KG/(3K+G)$.

Example 3

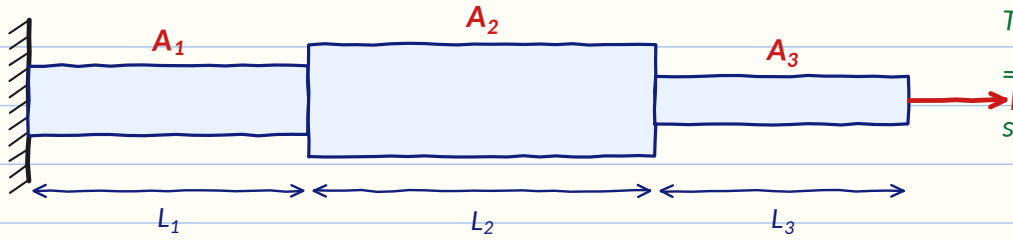
For a material $E = 2.1\times10^5$ N/mm², $G = 0.8\times10^5$ N/mm². Find μ and K.

$\mu = E/2G - 1 = 2.1/(2\times0.8) - 1 =$ **0.3125**

$K = E/[3(1-2\mu)] = 2.1\times10^5/[3(1-0.625)] =$ **1.87×10^5 N/mm²**

1.4 Bars of Varying Section

Load P acts on a bar whose area changes in steps. Each portion carries the same internal force (if loads act only at the ends).



Total $\Delta L = \Delta L_1 + \Delta L_2 + \Delta L$

$= \sum P L / (A E)$

same $P \Rightarrow \sigma_n = P/A_n$

$$\Delta L = \frac{P}{E}\left(\frac{L_1}{A_1} + \frac{L_2}{A_2} + \frac{L_3}{A_3}\right)$$

- If different E for each part: $\Delta L = P \sum (L_n)/(A_n E_n)$.
- Stress in each part is different: $\sigma_n = P/A_n$ (max stress in the smallest area).

Bar with several axial loads

- Draw a free-body diagram, split the bar at every load point.
- Find the axial force in each segment (tension +, compression –) starting from the free end.
- Total elongation = algebraic sum of the changes of each segment.

Example 4

A stepped bar: AB = 400 mm (A = 600 mm^2), BC = 300 mm (A = 400 mm^2), CD = 200 mm (A = 200 mm^2); axial pull 20 kN. E = 2×10^5 N/mm^2. Find total elongation.

$\Delta L = P/E(400/600 + 300/400 + 200/200)$

$= 20000/(2\times10^5)\ (0.667 + 0.75 + 1.0)$

$= 0.1 \times 2.417 =$ **0.2417 mm**

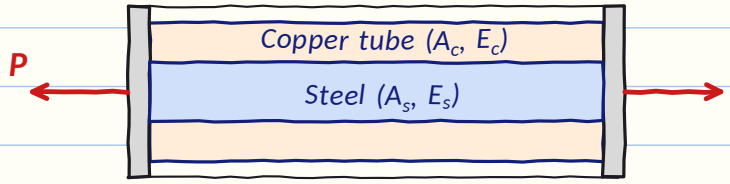
Tapering bar (circular, diameters $d_1 > d_2$)

$$\delta L = \frac{4 P L}{\pi E d_1 d_2}$$

- Rectangular taper (thickness t, width $b_1 \to b_2$): $\delta L = P L/(E t (b_1 - b_2)) \ln (b_1)/(b_2)$.

Temperature stresses

- A free bar heated by t°C expands $\delta L = \alpha L t$ – no stress.
- If expansion is prevented : $\sigma = \alpha t E$ (compressive on heating).
- Thermal load $P = \alpha t E A$. If support yields by δ: $\sigma = E(\alpha t - \delta/L)$.



Equilibrium : $P = P_s + P_c$

Compatibility : $\delta_s = \delta_c \Rightarrow \varepsilon_s = \varepsilon_c$

$\sigma_s / E_s = \sigma_c / E_c \Rightarrow \sigma_s = (E_s/E_c)\ \sigma_c$

$\sigma_s = m\ \sigma_c$, $m = E_s/E_c$ (modular ratio)

1.5 Composite (Compound) Bars

Two or more materials rigidly joined so that they deform together under load.

- Equilibrium : $P = P_1 + P_2 = \sigma_1 A_1 + \sigma_2 A_2$
- Compatibility : equal strain (same elongation) : $\varepsilon_1 = \varepsilon_2$

$$\frac{\sigma_1}{E_1} = \frac{\sigma_2}{E_2} \Rightarrow \sigma_1 = \frac{E_1}{E_2}\sigma_2$$

- Modular ratio $m = E_1/E_2$ (stiffer / softer).
- Load shared : $P_1 = P\,(A_1 E_1)/(A_{1E1} + A_{2E2})$, $P_2 = P\,(A_2 E_2)/(A_{1E1} + A_{2E2})$
- Equivalent area (in material 2): $A_{eq} = A_2 + m A_1$.

Example 5

Steel rod 20 mm dia. fits inside a copper tube 30 mm ext., 20 mm int. dia. Load 100 kN. $E_s = 2\times10^5$, $E_c = 1\times10^5$ N/mm^2. Find stresses.

$A_s = \pi/4(20)^2 = 314.2$ mm^2 ; $A_c = \pi/4(30^2 - 20^2) = 392.7$ mm^2

$\sigma_s = 2\,\sigma_c$ $(E_s/E_c = 2)$

$P = \sigma_s A_s + \sigma_c A_c = \sigma_c\,(2\times314.2 + 392.7) = 1021.1\,\sigma_c$

$\sigma_c = 100000/1021.1 =$ **97.93 N/mm^2**

$\sigma_s =$ **195.86 N/mm^2**

Temperature stresses in composite bars

- Different α : on heating, the material with higher α is restrained by the other ⇒ it is in compression, the other in tension.
- Equilibrium (no external load): $\sigma_1 A_1 = \sigma_2 A_2$ (equal & opposite forces).
- Compatibility: free expansion difference = elastic strains:

$$(\alpha_1 - \alpha_2)\,t = \frac{\sigma_1}{E_1} + \frac{\sigma_2}{E_2}$$

1.6 Shear, Bearing and Punching Stresses

- Shear stress acts **parallel** to the cross-section : $\tau = P/(A_{shear})$.
- Complementary shear : a shear stress on a plane is always accompanied by an equal shear stress of opposite sense on the perpendicular plane : $\tau_{xy} = \tau_{yx}$.
- Shear modulus $G = \tau/\varphi$ (φ = shear strain in radian).

Case	Shear area	τ
Single shear (rivet/bolt)	$\pi d^2/4$	$P / (\pi d^2/4)$
Double shear (clevis)	$2 \times \pi d^2/4$	$P / (2 \pi d^2/4)$
Punching a hole	$\pi d t$	$P / (\pi d t)$
Bearing (crushing)	$d \times t$	$P / (d t)$

Example 6 (Punching)

Force needed to punch a 25 mm hole in a 10 mm thick plate, shear strength τ_u = 300 N/mm^2.

Shear area = $\pi d t = \pi \times 25 \times 10 = 785.4$ mm^2

$P = \tau_u \times$ area = $300 \times 785.4 = 235\,619$ N ≈ **235.6 kN**

Example 7 (Double shear)

A 20 mm bolt in double shear carries 50 kN.

$\tau = P / (2A) = 50\,000 / (2 \times 314.16) =$ **79.58 N/mm^2**

1.7 Statically Indeterminate Bars and Temperature

Bar with intermediate loads (Example 8)

Uniform bar ABCD ($A = 500$ mm^2, $E = 2\times10^5$ N/mm^2) fixed at A. AB = 600, BC = 800, CD = 500 mm. Axial loads : 20 kN pull at D, 50 kN pull at C, 30 kN push at B.

Axial forces : CD = +20 kN ; BC = 20 + 50 = +70 kN ; AB = 70 – 30 = +40 kN

$\Delta L = 1/AE\ (20\times500 + 70\times800 + 40\times600)$ kN·mm $= (9\times10^7)/(500 \times 2\times10^5)$

$\Delta L =$ **0.9 mm** (elongation)

Bar fixed at both ends, load P at C (a from A, b from B)

- Equilibrium : $R_A + R_B = P$. Compatibility : total change of length = 0.
- AC extends under R_A ; CB shortens under R_B : $(R_A\ a)/AE = (R_B\ b)/AE$

$$R_A = \frac{P\ b}{L} \qquad R_B = \frac{P\ a}{L}$$

Example 9

Bar of 1.2 m between rigid walls, 60 kN axial load at 0.4 m from A ($a = 0.4$, $b = 0.8$).

$R_A = 60 \times 0.8 / 1.2 =$ **40 kN** ; $R_B = 60 \times 0.4 / 1.2 =$ **20 kN**

Example 10 (Thermal – rail)

12 m steel rail laid at 15°C, max temperature 45°C. $\alpha = 12\times10^{-6}$ /°C, $E = 2\times10^5$ N/mm^2.

Free expansion = $\alpha L t = 12\times10^{-6} \times 12000 \times 30 =$ **4.32 mm**

If ends fully restrained : $\sigma = \alpha\, t\, E = 12\times10^{-6} \times 30 \times 2\times10^{5} =$ **72 N/mm^2**

If 3 mm gap is left : $\sigma = E\,(\alpha\, t - \delta/L) = 2\times10^{5}(3.6\times10^{-4} - 2.5\times10^{-4}) =$ **22 N/mm^2**

1.8 More Solved Problems

Example 11 (Volumetric strain)

Bar 50 mm × 50 mm × 1000 mm, axial pull 100 kN, $E = 2\times10^5$ N/mm², $\mu = 0.3$. Find change in length and volume.

$\sigma = 100\,000 / 2500 = 40$ N/mm² ; $\varepsilon = 40 / 2\times10^5 = 2\times10^{-4}$

$\delta L = \varepsilon L =$ **0.2 mm**

$\varepsilon_v = \varepsilon (1 - 2\mu) = 2\times10^{-4} \times 0.4 = 8\times10^{-5}$

$\delta V = \varepsilon_v \times V = 8\times10^{-5} \times 2.5\times10^6 =$ **200 mm³**

Example 12 (Tapering circular bar)

Bar tapers uniformly from 40 mm to 20 mm dia. over 500 mm. P = 50 kN, $E = 2\times10^5$ N/mm².

$\delta L = 4 P L/(\pi E d_1 d_2) = (4 \times 50000 \times 500)/(\pi \times 2\times10^5 \times 40 \times 20)$

$\delta L = 10^8 / 5.027\times10^8 =$ **0.199 mm**

Example 13 (Composite bar with temperature)

Steel ($A = 500$ mm²) and copper ($A = 1000$ mm²) bars rigidly joined, heated by 60°C. $\alpha_s = 12\times10^{-6}$, $\alpha_c = 17\times10^{-6}$, $E_s = 2\times10^5$, $E_c = 1\times10^5$ N/mm².

Equilibrium : $\sigma_s A_s = \sigma_c A_c \Rightarrow \sigma_s = 2\,\sigma_c$

Compatibility : $(\sigma_s)/(E_s) + (\sigma_c)/(E_c) = (\alpha_c - \alpha_s)\, t = 5\times10^{-6} \times 60 = 3\times10^{-4}$

$\sigma_c\,(1\times10^{-5} + 1\times10^{-5}) = 3\times10^{-4} \Rightarrow \sigma_c =$ **15 N/mm²** (compression)

$\sigma_s =$ **30 N/mm²** (tension) ; check : 30×500 = 15×1000 = 15 kN ✓

Example 14 (Self-weight)

Vertical steel rod 30 m long, density 78.5 kN/m³, $E = 2\times10^5$ N/mm² hangs freely.

Max stress at top = $w L = 78.5\times10^{-6}$ N/mm³ × 30000 mm = **2.355 N/mm²**

$\delta L = (w L^2)/2E = (78.5\times10^{-6} \times 30000^2)/(2 \times 2\times10^5) =$ **0.177 mm**

1.9 Important Short Questions (Exam Oriented)

- Q1. State Hooke's law. A: $\sigma \propto \varepsilon$ within elastic limit; $\sigma = E\varepsilon$.
- Q2. Define Poisson's ratio. A: ratio of lateral strain to longitudinal strain; 0.25–0.33 for metals.
- Q3. What is factor of safety? A: ultimate (yield) stress / working stress.
- Q4. Write the relation among E, G, K. A: $E = 2G(1+\mu) = 3K(1-2\mu) = 9KG/(3K+G)$.
- Q5. What is modular ratio? A: $m = E_1 / E_2$ used for composite bars.
- Q6. Define proof stress. A: stress producing a specified permanent strain (0.2 %) in materials with no clear yield point.

- Q7. What is complementary shear stress? A: equal & opposite shear stress on perpendicular planes.
- Q8. Why are temperature stresses zero in a freely expanding bar? A: no restraint ⇒ no strain energy / no elastic strain.

Quick revision

- $\sigma = P/A$; $\varepsilon = \delta L/L$; $E = \sigma/\varepsilon$; $\delta L = PL/AE$
- $E = 2G(1+\mu) = 3K(1-2\mu) = 9KG/(3K+G)$
- Composite bar : same ε ; $\sigma_1/E_1 = \sigma_2/E_2$; $\sum P = P$
- Thermal : $\sigma = \alpha\, t\, E$ (restrained)

STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – II

Shear Force and Bending Moment

Handwritten Study Notes

Syllabus topics covered in this unit :

Definition of beam — Types of beams — Concept of shear force and bending moment — Point of contra flexure — Relation between S.F., B.M. and rate of loading — S.F. and B.M. diagrams for cantilever, simply supported and overhanging beams subjected to point loads, U.D.L., U.V.L., partial U.D.L., couple and combinations

Contents of these notes :

→ 2.1 Beams – Definition and Types
→ 2.2 Shear Force and Bending Moment
→ 2.3 Cantilever Beams
→ 2.4 Simply Supported Beams
→ 2.5 Overhanging Beams

2.1 Beams – Definition and Types

A beam is a structural member, usually horizontal, that carries loads transverse (perpendicular) to its axis and resists them by bending and shear.

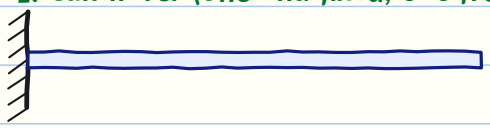
- Loads cause internal shear force (S.F.) and bending moment (B.M.) – these decide the section size.

Types of supports

- **Simple / roller support** : gives one vertical reaction ; beam free to rotate and slide.
- **Hinged (pin) support** : gives vertical + horizontal reaction ; rotation free.
- **Fixed support** : gives V, H reactions and a fixing moment ; no rotation or translation.

Types of beams

1. Cantilever (one end fixed, one free)



2. Simply supported



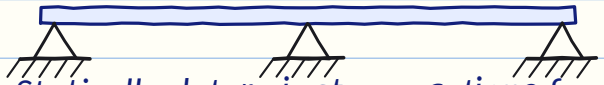
3. Overhanging (one/both ends)



4. Fixed (built-in) beam



5. Continuous beam (3 or more supports)



Propped cantilever = fixed + prop at free end

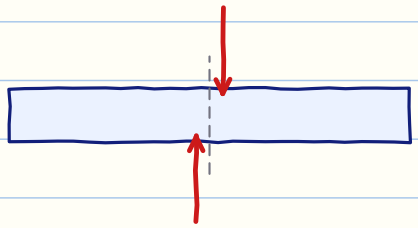
Determinate: 1, 2, 3 (statics only)

- Statically determinate : reactions found by $\Sigma V = 0$, $\Sigma H = 0$, $\Sigma M = 0$ (cantilever, simply supported, overhanging).
- Statically indeterminate : extra reactions – fixed, propped cantilever, continuous beams.

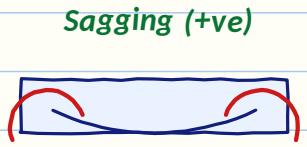
Types of loads

- Point (concentrated) load P – kN, acts at a point.
- Uniformly distributed load (U.D.L.) w – kN/m ; equivalent point load $W = w\,L$ at the centre.
- Uniformly varying load (U.V.L.) – intensity varies linearly ; triangular load $W = \frac{1}{2}\,w\,L$ acting at $L/3$ from the larger end.
- Couple / moment M – kN·m, produces rotation only.

+ve Shear : left part ↑, right part ↓

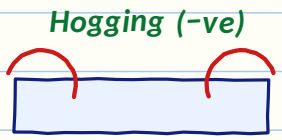


(clockwise shear on the segment)



Sagging (+ve)

concave upward



Hogging (−ve)

concave downward

2.2 Shear Force and Bending Moment

Shear force at a section = algebraic sum of all vertical forces to ONE side of the section.

Bending moment at a section = algebraic sum of moments of all forces to ONE side of the section about that section.

Sign conventions (sagging +ve)

- S.F. : +ve if the left part tends to move UP (and right part DOWN) – clockwise shear.
- B.M. : +ve (sagging) when it bends the beam concave upward ; −ve (hogging) when concave downward.
- Quick rule : forces/moments to the LEFT : upward force +ve, clockwise moment +ve ; to the RIGHT : reverse.

Relation between load, S.F. and B.M.

Consider an element dx with load intensity w (down) :

$$\frac{dF}{dx} = -w \qquad \frac{dM}{dx} = F$$

$$\frac{d^2M}{dx^2} = -w$$

- Slope of S.F. diagram = −(load intensity).
- Slope of B.M. diagram = S.F. at that section.
- Change in S.F. between two sections = − area of load diagram.
- Change in B.M. between two sections = area of S.F. diagram.
- Maximum B.M. occurs where S.F. = 0 (or changes sign).

Point of contra-flexure (P.C.)

The point where the bending moment changes sign (B.M. = 0 and curvature reverses). Occurs only in overhanging / fixed / continuous beams.

Shapes of diagrams

Loading on span	S.F.D.	B.M.D.
No load	horizontal line	inclined line
Point load	sudden jump = load	kink (slope changes)
U.D.L.	inclined line	parabola
U.V.L.	parabola	cubic parabola
Couple	no change	sudden jump = couple

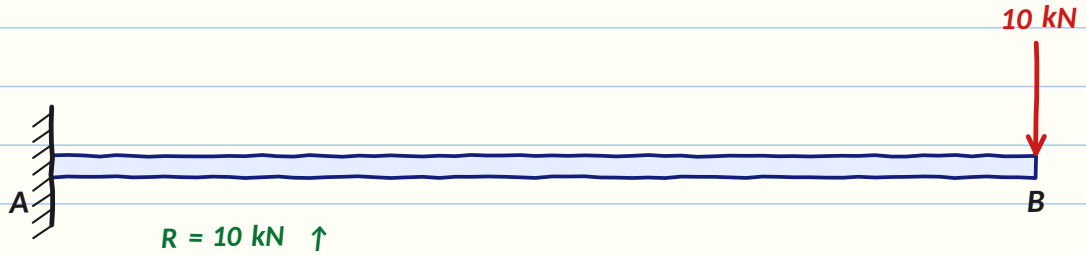
Procedure for drawing S.F.D and B.M.D

- 1. Replace distributed loads by equivalent point loads ; find reactions.
- 2. Divide the beam into segments at loads/supports.
- 3. Find F and M just left/right of every key section ; plot to scale.
- 4. Join using the shape rules ; locate F = 0 for M_{max} ; find P.C.

2.3 Cantilever Beams

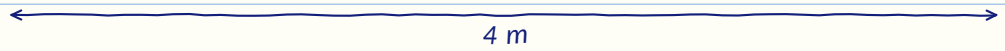
Free-end at B, fixed at A. Take x from the free end – no need to find reactions.

Case 1 : Point load W at free end (L = 4 m, W = 10 kN)

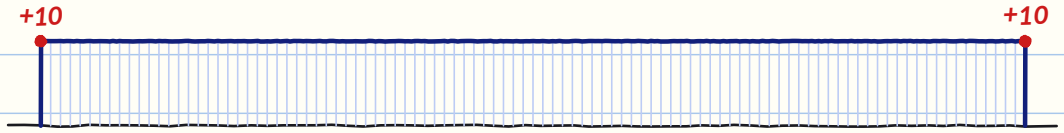


$R = 10$ kN ↑

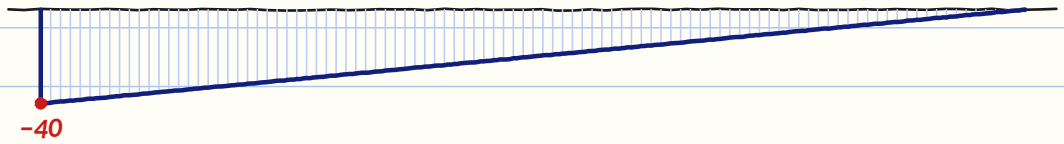
$M_A = 40$ kN·m (fixed-end moment)



S.F.D

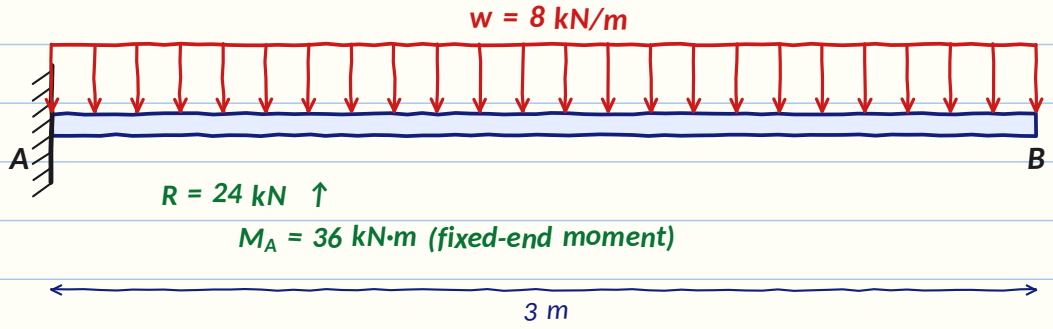


B.M.D

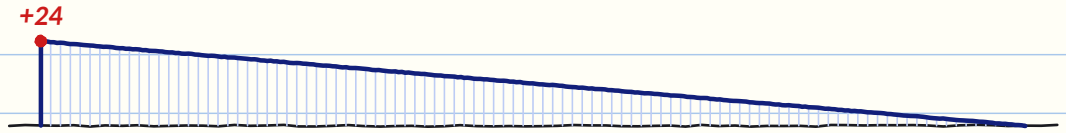


- $F_x = +W$ (constant) $= +10$ kN ; $M_x = -Wx$; $M_A = -WL = -10 \times 4 =$ **–40 kN·m**

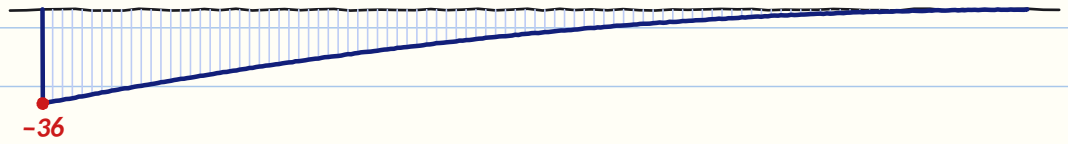
Case 2 : U.D.L. w over full span (L = 3 m, w = 8 kN/m)



S.F.D



B.M.D



- $F_x = w\,x$: $F_B = 0$, $F_A = w\,L$ = **24 kN** (straight line)
- $M_x = -\,w\,x^2/2$: $M_A = -\,8 \times 9/2$ = **–36 kN·m** (parabola)

Case 3 : U.V.L. – zero at free end, w at fixed end

- Total load $W = ½\,w\,L$ (acts at L/3 from the fixed end).

$$F_A = \frac{w\,L}{2} \qquad M_A = -\frac{w\,L^2}{6}$$

- $F_x = w\,x^2/2L$ (parabola) ; $M_x = -\,w\,x^3/6L$ (cubic).

Example 1 (Combination)

Cantilever 3 m : 10 kN at free end B, U.D.L. 4 kN/m over the whole span, couple 6 kN·m (clockwise) at mid-span.

F_B = 10 ; at mid : 10 + 4×1.5 = 16 kN ; at A : 10 + 4×3 = **22 kN**

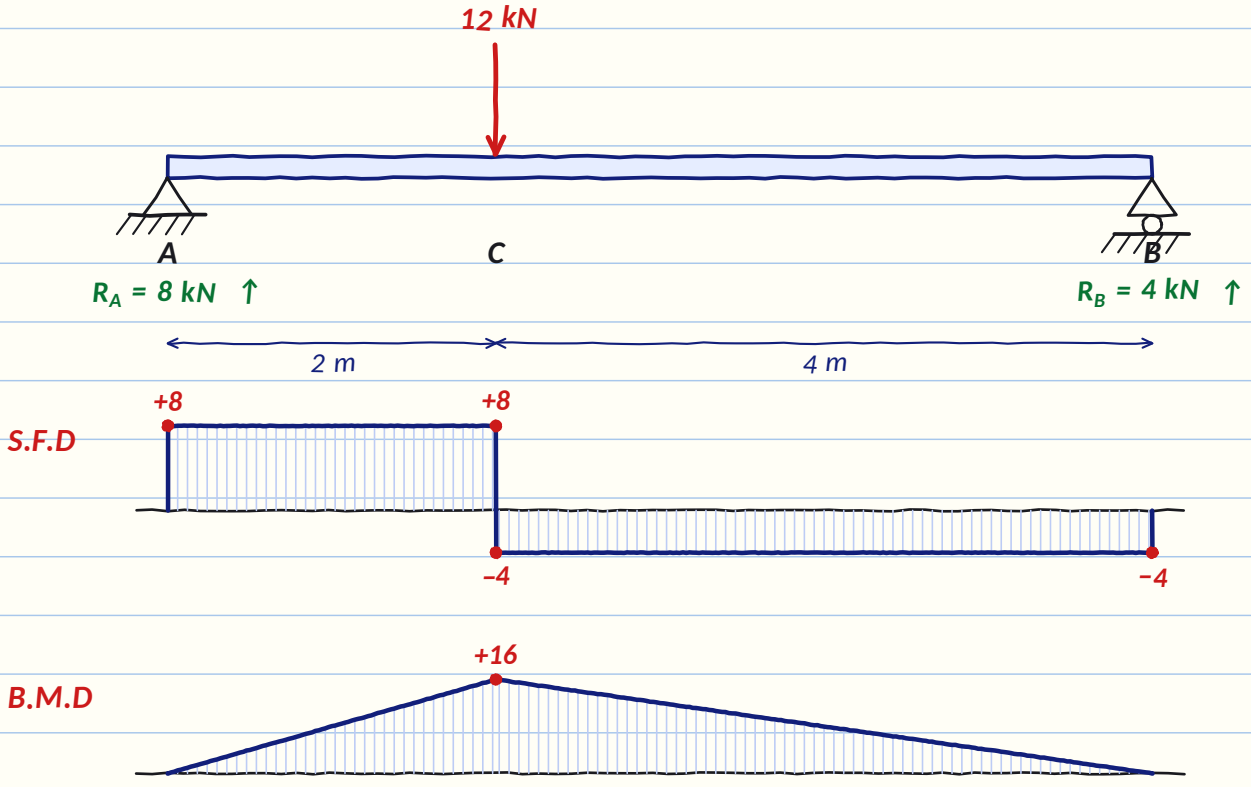
M just right of mid (from free end) = – (10×1.5 + 4×1.5×0.75) = – 19.5 kN·m

just left of mid = – 19.5 – 6 = **– 25.5 kN·m** (a clockwise couple on the right part is –ve)

M_A = – (10×3 + 4×3×1.5 + 6) = **– 54 kN·m** (hogging)

2.4 Simply Supported Beams

Case 4 : Point load (L = 6 m, W = 12 kN at C, 2 m from A)



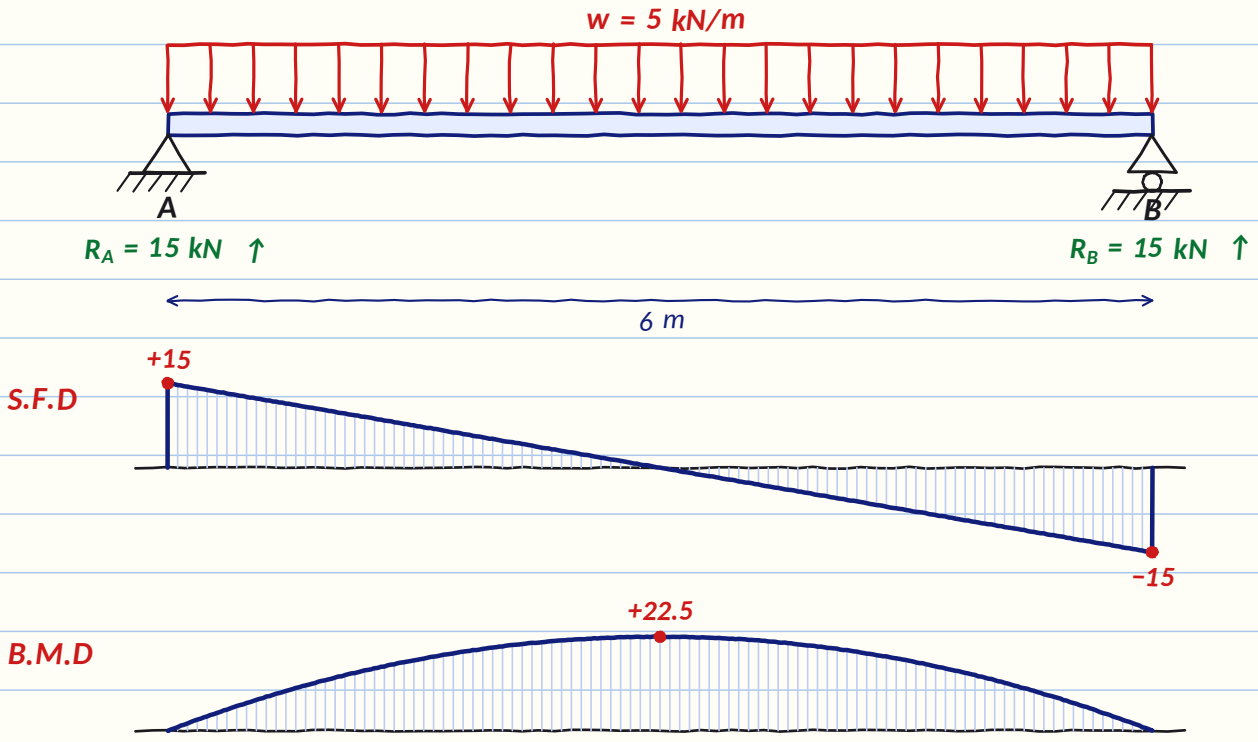
$R_B = 12 \times 2 / 6 =$ **4 kN** ; $R_A = 12 - 4 =$ **8 kN**

F : A to C = +8 ; C to B = 8 – 12 = –4 kN

$M_C = 8 \times 2 =$ **16 kN·m** (max, sagging) ; $M_A = M_B = 0$

- Central point load : $R = W/2$; $M_{max} = W\,L/4$.

Case 5 : U.D.L. over full span (w = 5 kN/m, L = 6 m)

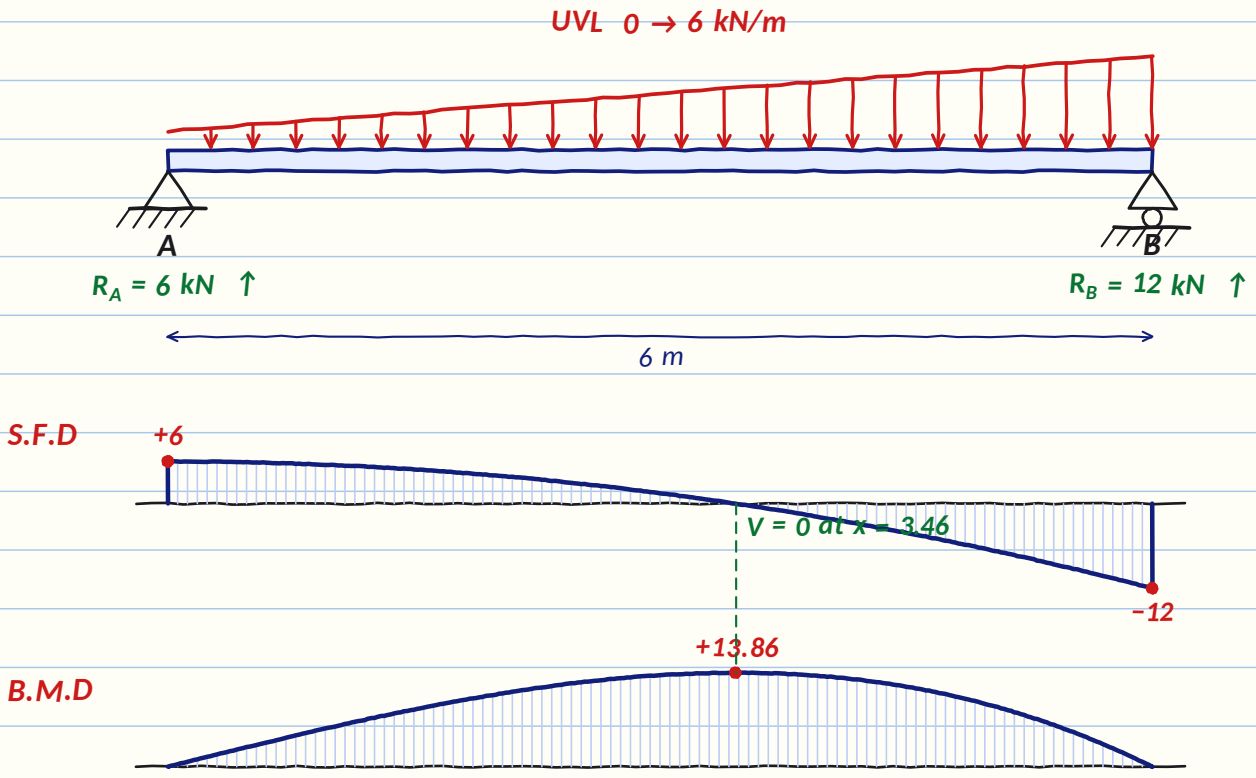


$R_A = R_B = w\,L/2 = 5 \times 6/2 =$ **15 kN**

$F_x = 15 - 5x$; $F = 0$ at $x = 3$ m (centre)

$M_{max} = w\,L^2/8 = 5 \times 36/8 =$ **22.5 kN·m** at centre (parabola)

Case 6 : U.V.L. (0 at A to 6 kN/m at B , L = 6 m)



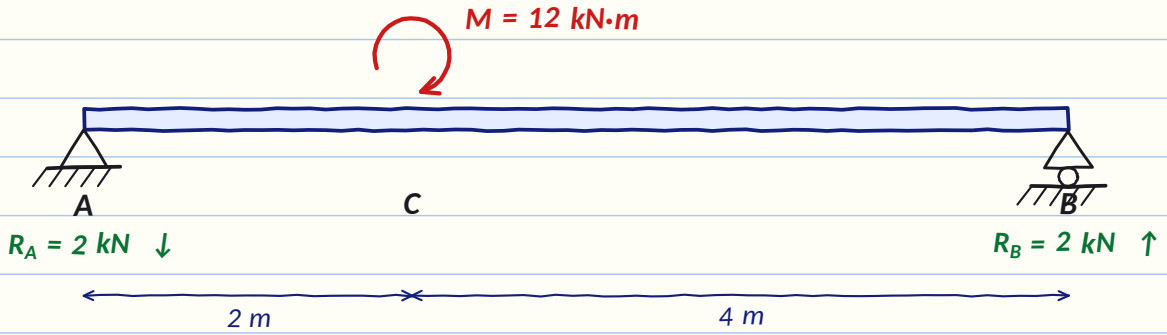
Total W = ½ × 6 × 6 = 18 kN ; $R_A = W/3$ = **6 kN** ; $R_B = 2W/3$ = **12 kN**

$F_x = 6 - x^2/2$; F = 0 at $x = L/\sqrt{3}$ = **3.464 m** from A

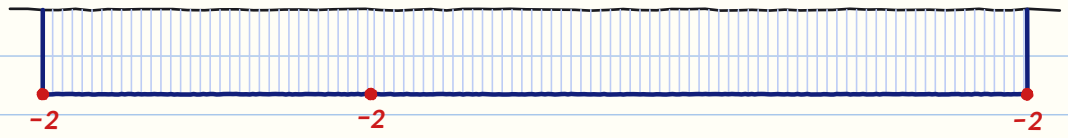
$M_x = 6x - x^3/6$; $M_{max} = 6(3.464) - 3.464^3/6$ = **13.86 kN·m**

- General : $M_{max} = 0.0642\ w\ L^2$ at 0.577 L from the thin end.

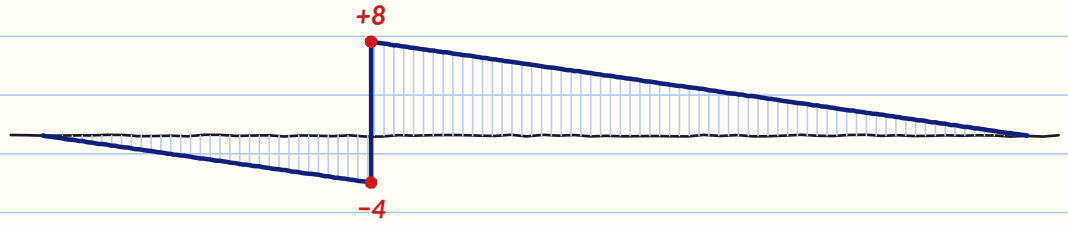
Case 7 : Couple on a simply supported beam (12 kN·m clockwise at C, 2 m fr



S.F.D



B.M.D



Take moments about A : $R_B \times 6 = 12 \Rightarrow R_B = 2$ kN (↑) ; $R_A = -2$ kN (i.e. 2 kN ↓)

S.F. = − 2 kN throughout (constant)

M just left of C = − 2 × 2 = **−4 kN·m** ; just right = −4 + 12 = **+8 kN·m**

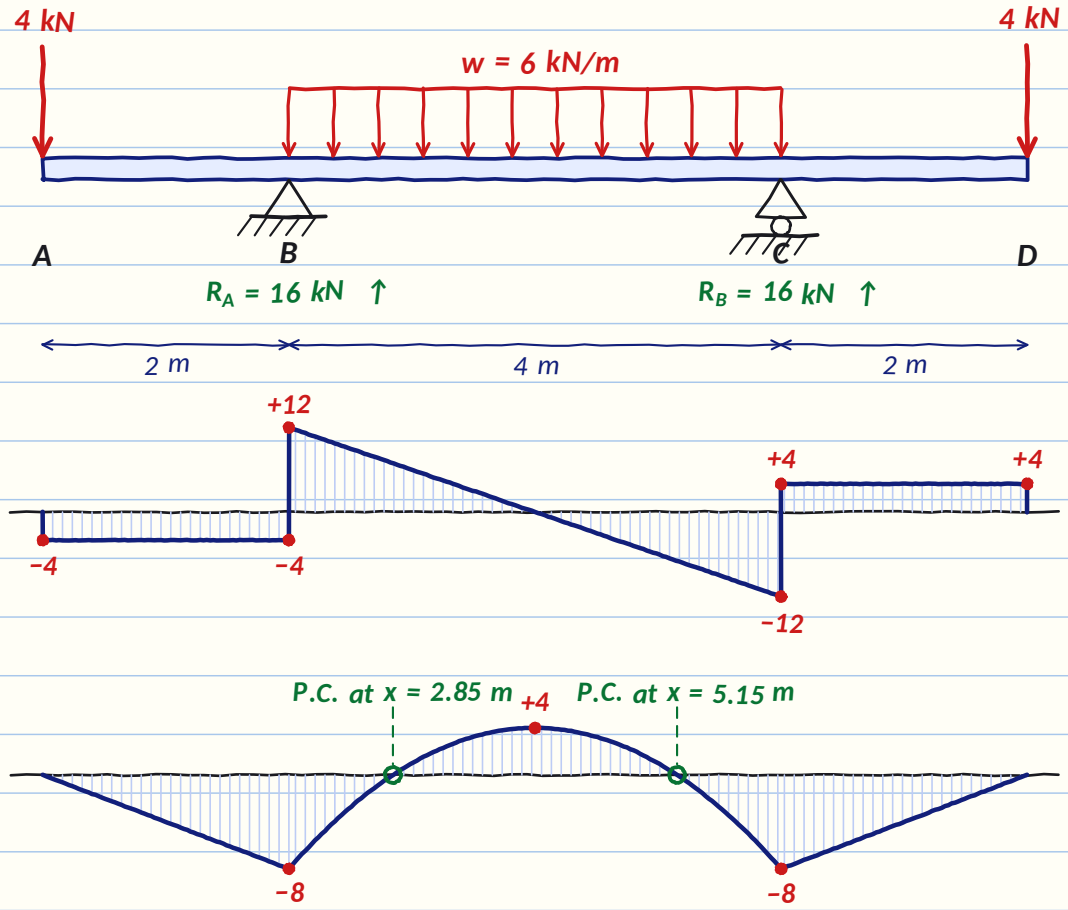
- A couple does not affect S.F. but makes the B.M. jump by its magnitude.

2.5 Overhanging Beams

Beam rests on two supports and extends beyond one or both. The B.M. changes sign ⇒ **points of contra-flexure**.

Case 8 : Overhang on both sides

Supports at B (2 m from A) and C (6 m from A) ; 4 kN at each free end A, D ; U.D.L. 6 kN/m between B and C.



By symmetry : $R_B = R_C = (4 + 4 + 24)/2 =$ **16 kN**

S.F. : A–B = −4 ; B right = −4 + 16 = +12 ; at C left = 12 – 24 = −12 ; C right = −12 + 16 = +4 ; C–D = +4

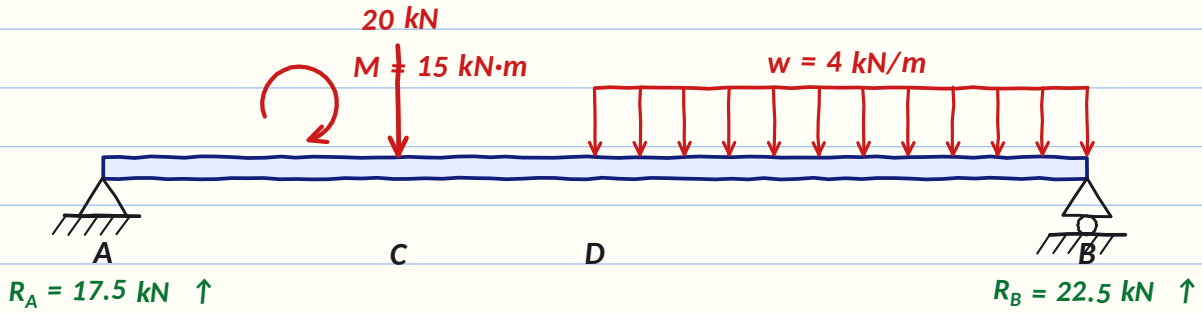
B.M. : $M_B = -4 \times 2 =$ **−8 kN·m** ; $M_C = -4 \times 2 = -8$ kN·m

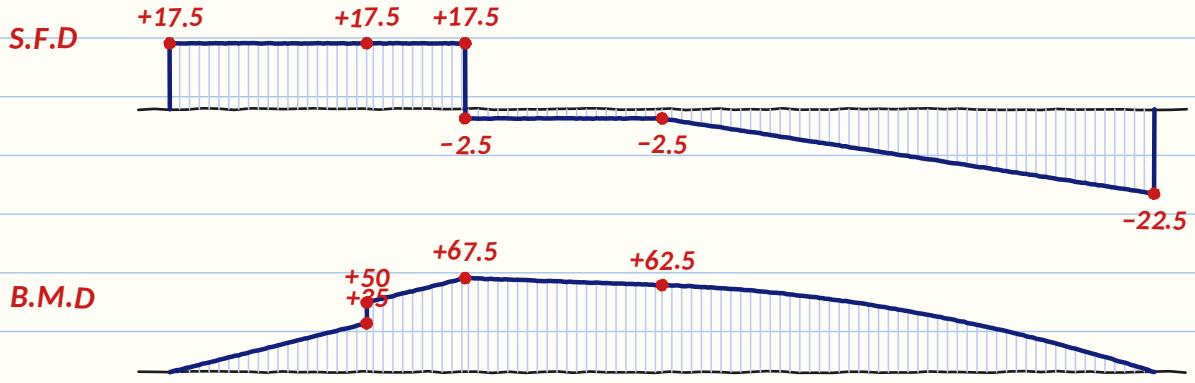
F = 0 at 2 m right of B ⇒ $M = -8 + 12\times2 - 6\times2^2/2 =$ **+4 kN·m** (max sagging, midspan)

Contra-flexure : $-8 + 12s - 3s^2 = 0 \Rightarrow s = 0.845$ m or 3.155 m from B ⇒ **x = 2.845 m and 5.155 m from A**

Case 9 : Combination (point load + partial U.D.L. + couple)

SS beam 10 m : 20 kN at 3 m from A, U.D.L. 4 kN/m from 5 m to B (10 m), clockwise couple 15 kN·m at 2 m from A.





Moments about A : $R_B \times 10 = 20\times3 + (4\times5)\times7.5 + 15 = 60 + 150 + 15 = 225$

R_B = **22.5 kN** ; $R_A = 40 - 22.5 =$ **17.5 kN**

S.F. : A–C(2) 17.5 ; 2–3 : 17.5 ; right of P : −2.5 ; at 5 m : −2.5 ; at B : −2.5 − 20 = −22.5

B.M. : $M_{x=2}$ left = 35 ; right = 35 + 15 = 50 ; $M_{x=3} = 17.5\times3 + 15 =$ **67.5 kN·m** (max, S.F. changes sign)

$M_{x=5} = 17.5\times5 + 15 - 20\times2 = 62.5$; $M_B = 0$ (check ✓)

Standard results – quick table

Beam / load	Max S.F.	Max B.M.
Cantilever, W at end	W	W L
Cantilever, w/unit length	w L	w $L^2/2$
Cantilever, U.V.L. (0 at free end)	w L/2	w $L^2/6$
S.S., W at centre	W/2	W L/4
S.S., W at distance a from A	W b/L	W a b/L
S.S., U.D.L.	w L/2	w $L^2/8$
S.S., U.V.L. (0→w)	w L/3	0.0642 w L^2

Important points to remember

- Draw S.F.D and B.M.D below the loaded beam, same scale for length.
- Always check: sum of reactions = total load ; M at free/pinned end = 0.
- Max B.M. lies at F = 0 – equate $F_x = 0$ to get its location.
- For an overhanging beam, M changes sign – find P.C. by $M_x = 0$.
- Mark +/– signs, units and key values on every diagram.

STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – III

Flexural and Shear Stresses

Handwritten Study Notes

Syllabus topics covered in this unit :

Theory of simple bending — Assumptions — Bending equation — Neutral axis — Bending stresses — Section modulus of rectangular, circular (solid & hollow), I, T, Angle and Channel sections — Design of simple beams — Shear stress formula and distribution (rectangular, circular, I, T, Angle) — Torsion of circular shafts

Contents of these notes :

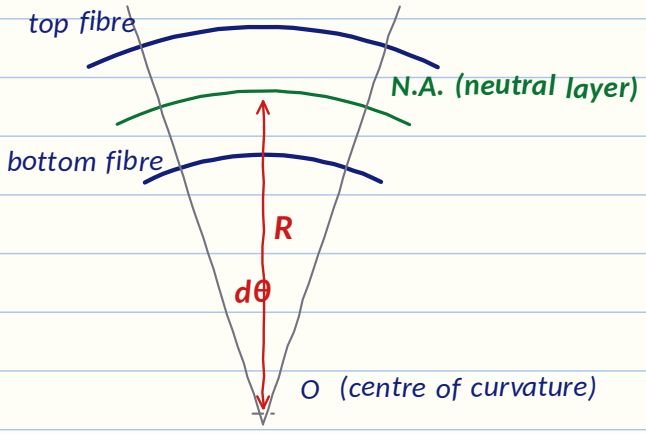
→ 3.1 Theory of Simple Bending
→ 3.2 Section Modulus of Common Sections
→ 3.3 Bending Stress Problems and Design
→ 3.4 Shear Stresses in Beams
→ 3.5 Torsion of Circular Shafts
→ 3.6 Derivations and More Solved Problems

3.1 Theory of Simple Bending

Pure (simple) bending : a beam segment subjected to a constant bending moment with zero shear force – only bending stresses (σ) act in it.

Assumptions

- Beam material is homogeneous and isotropic ; obeys Hooke's law (stress within elastic limit).
- E is same in tension and compression.
- Plane sections remain plane after bending (transverse sections stay ⊥ to the neutral surface).
- Beam is initially straight, of uniform section ; radius of curvature is large compared with depth.
- Loads act in the plane of symmetry (no twisting) ; each layer is free to expand/contract.
- Shear effects are neglected in pure bending.



Layer at distance y from N.A.:

original length = $R\,d\theta$

new length = $(R + y)\,d\theta$

strain $\varepsilon = y\,d\theta / (R\,d\theta) = y / R$

stress $\sigma = E\,\varepsilon = E\,y / R$

$\Rightarrow \quad \sigma / y = E / R$

Neutral axis (N.A.)

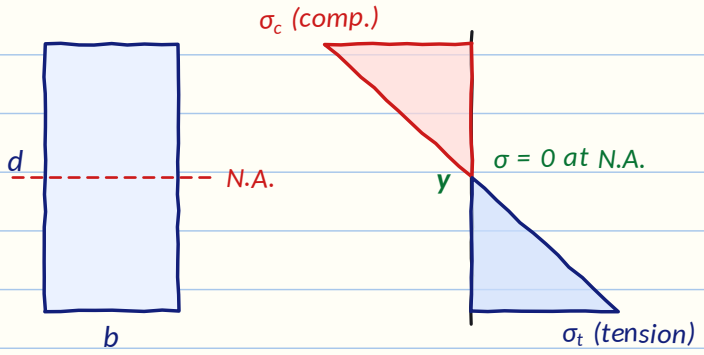
The layer that is neither stretched nor compressed is the neutral layer ; its trace on a cross-section is the neutral axis. N.A. passes through the centroid of the section.

- Proof : net axial force = $\int \sigma\,dA = (E/R) \int y\,dA = 0 \Rightarrow \int y\,dA = 0$ (centroidal axis).

Derivation of Bending Equation

- Layer at distance y from N.A. : strain $\varepsilon = y / R \Rightarrow \sigma = E\,y / R$.
- Moment of the force on strip dA about N.A. : $dM = \sigma\,dA \cdot y = (E/R)\,y^2\,dA$.
- Total $M = (E/R) \int y^2\,dA = (E/R)\,I$ where I = moment of inertia about N.A.

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$



σ varies linearly with y

N.A. passes through centroid

$\sigma_{max} = M\, y_{max} / I = M / Z$

$Z = I / y_{max}$ (section modulus)

Moment of resistance M = σ Z

- Bending stress varies linearly ; zero at N.A. ; max at the extreme fibre.

$$\sigma_{max} = \frac{M}{I}\, y_{max} = \frac{M}{Z} \qquad Z = \frac{I}{y_{max}}$$

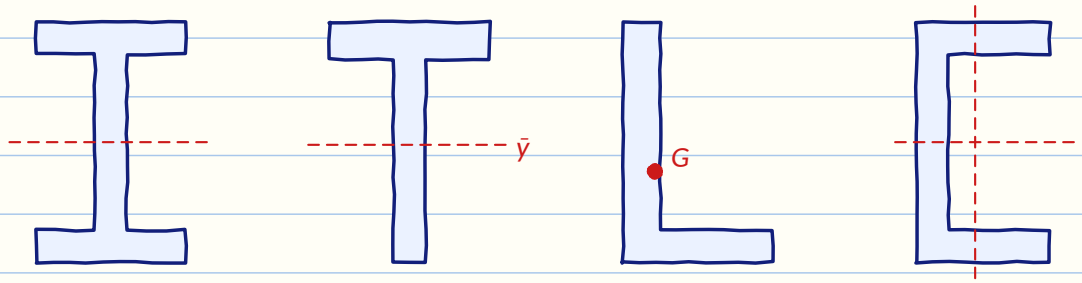
- Z = section modulus (mm^3) : a measure of the strength of the section in bending. Moment of resistance $M_r = \sigma_{permissible} \times Z$.

3.2 Section Modulus of Common Sections

- Parallel axis theorem : $I = I_G + A\,h^2$ (h = distance between parallel axes).
- Perpendicular axis theorem : $I_z = I_x + I_y$ (plane areas).

Section	I (about N.A.)	$Z = I / y_{max}$
Rectangle $b \times d$	$b\,d^3/12$	$b\,d^2/6$
Hollow rectangle (B×D, b×d)	$(B\,D^3 - b\,d^3)/12$	$(B\,D^3 - b\,d^3)/6D$
Solid circle dia. d	$\pi\,d^4/64$	$\pi\,d^3/32$
Hollow circle D, d	$\pi\,(D^4 - d^4)/64$	$\pi\,(D^4 - d^4)/32\,D$
Triangle (base b, ht h) about base	$b\,h^3/12$	—
Triangle about centroid	$b\,h^3/36$	$b\,h^2/24$ (apex side)

- For a given area, I and Z are larger when material is placed away from the N.A.
  ⇒ I, T, hollow sections are economical.



I-section T-section Angle Channel

Procedure for I, T, Angle, Channel

- 1. Split the section into simple rectangles : areas A_1, A_2, ...
- 2. Find centroid : $\bar{y} = (\sum A\,y) / \sum A$ (measure from one edge) ; similarly $\bar{x}$ for unsymmetrical sections.
- 3. $I_{NA} = \sum (I_G + A\,h^2)$ of each rectangle about the centroidal axis.
- 4. $Z_{top} = I / y_{top}$, $Z_{bottom} = I / y_{bottom}$; use the smaller Z for max stress.

Example 1 (T-section)

Flange 100 × 20 mm ; web 20 × 100 mm (overall depth 120 mm). Find $\bar{y}$, I and Z.

$A_1 = 2000$ ($y_1 = 10$ from top), $A_2 = 2000$ ($y_2 = 70$) ⇒ $\bar{y} = (2000\times10 + 2000\times70)/4000 =$ **40 mm from top**

$I = (100\times20^3/12 + 2000\times30^2) + (20\times100^3/12 + 2000\times30^2)$

$I = 66\,667 + 1.8\times10^6 + 1.667\times10^6 + 1.8\times10^6 =$ **5.333×10^6 mm^4**

$Z_{top} = I/40 =$ **1.333×10^5 mm^3** ; $Z_{bottom} = I/80 =$ **0.667×10^5 mm^3**

Example 2 (I-section)

Flanges 200 × 20 ; web 10 × 300 ; overall depth 340 mm.

$I = [200 \times 340^3 - (200 - 10) \times 300^3] / 12 = [7.8608\times10^9 - 5.13\times10^9]/12 =$ **2.2757×10^8 mm^4**

$Z = I / 170 =$ **1.3386×10^6 mm^3**

Example 3 (Angle 100 × 80 × 10)

$A_1 = 100\times10 = 1000$ $(x=5, y=50)$; $A_2 = 70\times10 = 700$ $(x=45, y=5)$

$\bar{x} = (5000 + 31500)/1700 = 21.47$ mm ; $\bar{y} = (50000 + 3500)/1700 = 31.47$ mm

$I_{xx} = (10\times100^3/12 + 1000\times18.53^2) + (70\times10^3/12 + 700\times26.47^2) \approx$ **1.673×10^6 mm^4**

Channel section

- Symmetrical about the horizontal axis : I_{xx} by the usual formula ; unsymmetrical about the vertical axis ⇒ find $\bar{x}$ from the back of the web ; $I_{yy} = \sum(I_G + A\,h^2)$ about $\bar{x}$.
- Loads should act through the shear centre of a channel, otherwise twisting occurs.

3.3 Bending Stress Problems and Design

Example 4

Rectangular beam 100 mm × 200 mm, simply supported over 4 m, U.D.L. 10 kN/m.

$M_{max} = w L^2/8 = 10 \times 16/8 = 20$ kN·m $= 20\times10^6$ N·mm

$I = 100 \times 200^3/12 = 66.67\times10^6$ mm^4

$\sigma_{max} = M\,y/I = 20\times10^6 \times 100 / 66.67\times10^6 =$ **30 N/mm^2**

Example 5 (Circular section)

Solid circular beam $d = 100$ mm, $M = 5$ kN·m.

$Z = \pi d^3/32 = 98\,175$ mm$^3 \Rightarrow \sigma = 5\times10^6 / 98\,175 =$ **50.9 N/mm^2**

Example 6 (Cantilever – extreme stresses)

Cantilever 2 m, point load 5 kN at the end, T-section of Example 1 with flange on top.

$M_A = 5 \times 2 = 10$ kN·m (hogging ⇒ top in tension)

σ_{top} (tension) $= M/Z_{top} = 10\times10^6 / 1.333\times10^5 =$ **75 N/mm^2**

σ_{bottom} (compression) $= 10\times10^6 / 0.667\times10^5 =$ **150 N/mm^2**

Design of simple beams

Procedure : (1) find M_{max} from the B.M.D. (2) $Z_{required} = M_{max} / \sigma_{permissible}$ (3) choose a section with $Z \geq Z_{req}$ (4) check shear stress and deflection.

Example 7 (Rectangular timber beam, depth = 2 × width)

S.S. beam 5 m, total load 12 kN/m (incl. self weight), $\sigma_{perm} = 150$ N/mm^2.

$M = 12 \times 25/8 = 37.5$ kN·m ; $Z_{req} = 37.5\times10^6/150 = 2.5\times10^5$ mm^3

$Z = b\,d^2/6 = b\,(2b)^2/6 = 2b^3/3 = 2.5\times10^5 \Rightarrow b = 72.1$ mm

Provide **75 mm × 150 mm** ($Z = 2.81\times10^5$ mm^3 > required ✓)

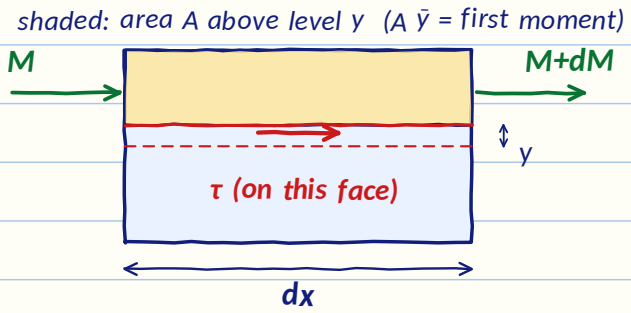
- For steel use I-sections : select the lightest rolled section from tables with $Z_{xx} \geq 250$ cm^3.

Strength comparison

- Beams of equal area : I-section ≫ square > circle for the same M (larger Z).
- Rectangle on edge vs flat : Z ratio = d/b (deep side vertical is stronger).

3.4 Shear Stresses in Beams

- A shear force F induces vertical shear stress τ on the cross-section and equal horizontal (complementary) shear stress on longitudinal planes.



Force on left face = $\int \sigma\, dA = (M/I)\, A\, \bar{y}$

Force on right face = $((M+dM)/I)\, A\, \bar{y}$

Difference = $(dM/I)\, A\, \bar{y}$

Balanced by $\tau\, b\, dx$:

$\tau\, b\, dx = (dM/I)\, A\, \bar{y}$

$\tau = F\, A\, \bar{y} / (I\, b)$ $(F = dM/dx)$

$$\tau = \frac{F\, A\, \bar{y}}{I\, b}$$

- F = shear force ; A = area of the part of section beyond the level considered ; $\bar{y}$ = distance of its centroid from N.A. ; b = width at that level ; I = M.I. of whole section about N.A.

Rectangular section (b × d)

- At height y from N.A. : $A\, \bar{y} = b\,(d/2 - y)(d/2 + y)/2 \Rightarrow \tau = F/2I((d^2)/4 - y^2)$ (parabolic).

$$\tau_{max}\ (\text{at N.A.}) = \frac{3F}{2\, b\, d} = 1.5\, \tau_{avg} \qquad \tau = 0 \text{ at top/bottom}$$

Circular section (radius R)

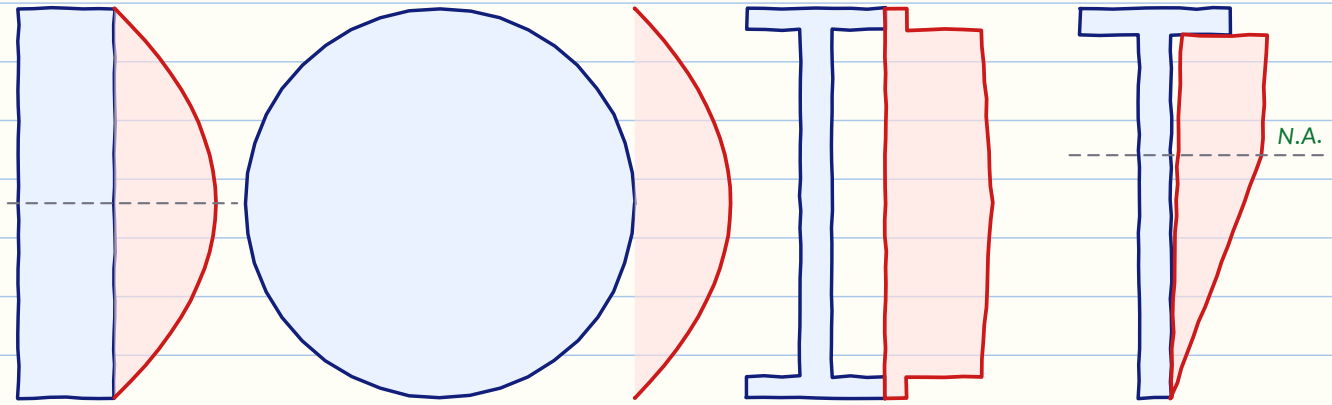
$$\tau_{max} = \frac{4F}{3\,\pi\, R^2} = \frac{4}{3}\, \tau_{avg} \quad (\text{at N.A.})$$

I-section

- Flange : τ small (wide b) ; at flange–web junction τ jumps (b changes from flange width to web thickness) ; max at N.A.
- Web carries about 95 % of the shear force, flanges carry bending ⇒ economical.

T-section and Angle

- T : same trend ; max at N.A. (not at the middle of depth) ; zero at free edges. Angle : max at N.A. of the leg ; shear centre lies at the corner.



Rectangle $\tau_{max} = 1.5\ \tau_{avg}$ | **Circle** $\tau_{max} = (4/3)\ \tau_{avg}$ | **I-section** web carries ≈ 95 % of F | **T-section** max at N.A.

Example 8 (Rectangular)

$b = 100$, $d = 200$ mm, $F = 30$ kN : $\tau_{avg} = 30000/20000 = 1.5 \Rightarrow \tau_{max} = 2.25$ N/mm^2

Example 9 (I-section of Example 2, F = 100 kN)

$A\bar{y}$ at N.A. $= 200\times20\times160 + 10\times150\times75 = 7.525\times10^5$ mm^3

$\tau_{NA} = 100000 \times 7.525\times10^5 / (2.2757\times10^8 \times 10) =$ **33.07 N/mm^2**

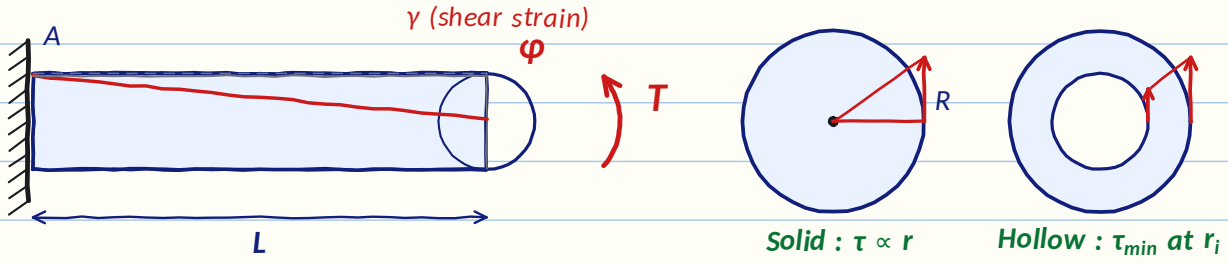
At web (junction) : $A\bar{y} = 6.4\times10^5 \Rightarrow \tau =$ **28.1 N/mm^2** ; in flange ($b = 200$) : $\tau =$ **1.41 N/mm^2**

3.5 Torsion of Circular Shafts

Torsion : a shaft is twisted by a torque T about its axis ⇒ shear stress on cross-sections.

Assumptions

- Material is homogeneous, isotropic, elastic (Hooke's law) ; shaft is circular and uniform.
- Plane cross-sections remain plane and radii remain straight after twisting.
- Twist is uniform along the length ; stress is within proportional limit.



Derivation of torsion equation

- Shear strain at radius r : $\gamma = r\,\varphi / L$; $\tau = G\,\gamma = G\,r\,\varphi / L$.
- Torque on ring dA at radius r : $dT = \tau\, r\, dA = (G\,\varphi / L)\, r^2\, dA \Rightarrow T = (G\,\varphi / L)\, J$.

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\,\varphi}{L}$$

- J = polar moment of inertia ; τ_{max} at outer surface (r = R).

Shaft	J	$Z_p = J/R$
Solid dia. d	$\pi\, d^4/32$	$\pi\, d^3/16$
Hollow D, d	$\pi\,(D^4 - d^4)/32$	$\pi\,(D^4 - d^4)/16\,D$

- Torsional rigidity = G J ; angle of twist $\varphi = T\,L / G\,J$ (radian).

$$\text{Power } P = \frac{2\,\pi\,N\,T}{60} \quad (T \text{ in N·m, } P \text{ in watt})$$

- Hollow shafts are stronger and lighter per kg than solid shafts (material near the axis is under-stressed).

Example 10 (Solid shaft)

$d = 50$ mm, $\tau_{perm} = 40$ N/mm^2, $N = 300$ rpm, $G = 80$ GPa, $L = 2$ m.

$T = \tau \pi d^3/16 = 40 \times 24\,544 = 9.817\times10^5$ N·mm = **0.982 kN·m**

$P = 2\pi N T / 60 = 2\pi \times 300 \times 981.7 / 60 =$ **30.84 kW**

$\varphi = T L / G J = 9.817\times10^5\times2000 / (8\times10^4 \times 6.136\times10^6) =$ **0.0040 rad = 0.229°**

Example 11 (Hollow shaft D = 60, d = 40 mm, T = 1 kN·m)

$J = \pi (60^4 - 40^4)/32 = 1.021\times10^6$ mm^4

$\tau_{max} = T R / J = 10^6 \times 30 / 1.021\times10^6 =$ **29.4 N/mm^2**

3.6 Derivations and More Solved Problems

Moment of inertia – by integration

- Rectangle ($b \times d$) about N.A. : strip of width b, thickness dy at distance y :

$$I = \int y^2\, b\, dy \quad (\text{from } -d/2 \text{ to } d/2) = \frac{b\,d^3}{12}$$

- Circle (radius R) : polar M.I. $J = \int r^2 (2\pi r\, dr) = (\pi R^4)/2 = (\pi d^4)/32$.
- By the perpendicular axis theorem $I_x = I_y = J/2 \Rightarrow$ **$I = \pi d^4/64$**.

Example 12 (Channel section)

Channel : web 10 × 200 mm, two flanges projecting 70 × 10 mm each (flange width 80, overall depth 200).

$A = 2000 + 2 \times 700 = 3400$ mm^2 ; $\bar{x} = (2000\times5 + 1400\times45)/3400 =$ **21.47 mm** from back of web

$I_{xx} = 10\times200^3/12 + 2\,[70\times10^3/12 + 700\times95^2] =$ **1.931×10^7 mm^4** $\Rightarrow Z_{xx} = 1.931\times10^5$ mm^3

$I_{yy} = [200\times10^3/12 + 2000\times16.47^2] + 2\,[10\times70^3/12 + 700\times23.53^2] \approx$ **1.906×10^6 mm^4**

Example 13 (Stress distribution in I-section, M = 100 kN·m)

Use the I-section of Example 2 ($I = 2.2757\times10^8$ mm^4, $Z = 1.3386\times10^6$ mm^3).

$\sigma_{max} = M/Z = 100\times10^6 / 1.3386\times10^6 =$ **74.7 N/mm^2** (top/bottom fibre)

At web–flange junction ($y = 150$) : $\sigma = 74.7 \times 150/170 =$ **65.9 N/mm^2**

Flanges alone : $I_f = (200\times340^3 - 200\times300^3)/12 = 2.05\times10^8 \Rightarrow$ about 90 % of I – so flanges resist most of the bending moment.

Example 14 (Cantilever with U.D.L. – check of rectangular section)

Cantilever 1.5 m, U.D.L. 20 kN/m, section 120 mm × 240 mm.

$M_{max} = 20 \times 1.5^2/2 = 22.5$ kN·m ; $Z = 120 \times 240^2/6 = 1.152\times10^6$ mm^3

$\sigma_{max} = 22.5\times10^6 / 1.152\times10^6 =$ **19.53 N/mm^2** ; $F_{max} = 30$ kN $\Rightarrow \tau_{max} = 1.5 \times$

30000/28800 = **1.56 N/mm²**

Example 15 (Shear stress in T-section, F = 20 kN)

Section of Example 1 : $I = 5.333\times10^6$ mm⁴, N.A. 40 mm from top.

At N.A. ($b = 20$) : $A\bar{y} = 2000\times30 + 20\times20\times10 = 64\,000$ mm³ ⇒ $\tau =$ $20000\times64000/(5.333\times10^6\times20)$ = **12.0 N/mm²**

Junction – web side ($b = 20$) : $A\bar{y} = 60\,000$ ⇒ **11.25 N/mm²** ; flange side ($b = 100$) : **2.25 N/mm²**

Hollow vs solid shaft (same torque, same τ_{max})

Solid $d = 100$ mm ; hollow shaft with $d_i / D_o = 0.6$.

$D_o^3 (1 - 0.6^4) = 100^3 \Rightarrow D_o = 104.7$ mm ; $d_i = 62.8$ mm

$A_{hollow} = \pi/4\,(104.7^2 - 62.8^2) = 5512$ mm² vs $A_{solid} = 7854$ mm² ⇒ **≈ 30 % saving in weight**

Example 16 (Safe U.D.L. on a timber beam)

S.S. beam 6 m, section 150 × 300 mm, $\sigma_{perm} = 10$ N/mm².

$Z = 150 \times 300^2/6 = 2.25\times10^6$ mm³ ; $M_r = \sigma Z = 22.5$ kN·m

$w L^2/8 = 22.5 \Rightarrow w = 22.5 \times 8 / 36 =$ **5 kN/m** (safe U.D.L., incl. self weight)

Example 17 (Hollow circular beam)

$D = 120$ mm, $d = 80$ mm, $M = 8$ kN·m.

$I = \pi\,(120^4 - 80^4)/64 = 8.168\times10^6$ mm⁴

$\sigma_{max} = M y/I = 8\times10^6 \times 60 / 8.168\times10^6 =$ **58.8 N/mm²**

Radius of curvature $R = E I / M = 2\times10^5 \times 8.168\times10^6 / 8\times10^6 =$ **204 m**

Example 18 (Diameter of a circular beam)

$M = 12$ kN·m, $\sigma_{perm} = 100$ N/mm².

$Z = M/\sigma = 1.2\times10^5$ mm³ $= \pi d^3/32 \Rightarrow d^3 = 1.222\times10^6 \Rightarrow$ **d = 107 mm → provide 110 mm**

Beam of uniform strength (concept)

- Section varies so that σ_{max} is the same at every section : M / Z = constant.
- Cantilever with end load : rectangular b constant ⇒ $d \propto \sqrt{x}$ (parabolic profile) ; d constant ⇒ $b \propto x$ (triangular plan).
- Saves material ; used in leaf springs.

Common mistakes to avoid

- Using the wrong y : y is always measured from the neutral axis (centroid), not from the bottom edge.
- For unsymmetrical sections compute stresses at BOTH extreme fibres : σ_t and σ_c

are different.

- Keep units consistent : N and mm (M in N·mm = kN·m × 10^6).

Shafts in series and parallel

- Series (different sections, same torque T) : total twist $\varphi = \varphi_1 + \varphi_2 = T\,L_1/G_{1J1} + T\,L_2/G_{2J2}$.
- Parallel (composite shaft, same twist) : $\varphi_1 = \varphi_2$ and $T = T_1 + T_2 \Rightarrow T_1/T_2 = G_1\,J_1 / G_2\,J_2$ (for equal lengths).

Formula summary – Unit III

Quantity	Formula
Bending equation	$M/I = \sigma/y = E/R$
Section modulus	$Z = I / y_{max}$
Shear stress	$\tau = F\,A\,\bar{y}\,/(I\,b)$
Rectangle τ_{max}	$3F/2bd = 1.5\,\tau_{avg}$
Circle τ_{max}	$4F/3A = 1.33\,\tau_{avg}$
Torsion	$T/J = \tau/r = G\,\varphi/L$
Power transmitted	$P = 2\pi\,N\,T\,/\,60$

Short questions

- Define section modulus. A: $Z = I / y_{max}$ (mm^3).
- Where is bending stress zero? A: at the neutral axis ; shear stress is zero at extreme fibres.
- Max shear stress / average : rectangle 1.5 ; solid circle 4/3.
- What is polar modulus? A: $Z_p = J / R$.

STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – IV

Deflection of Beams

Handwritten Study Notes

Syllabus topics covered in this unit :

Double integration and Macaulay's methods — Slope and deflection for cantilever, simply supported and overhanging beams under point loads, U.D.L., U.V.L., partial U.D.L., couple and combinations — Mohr's theorems — Moment area method — application to cantilevers

Contents of these notes :

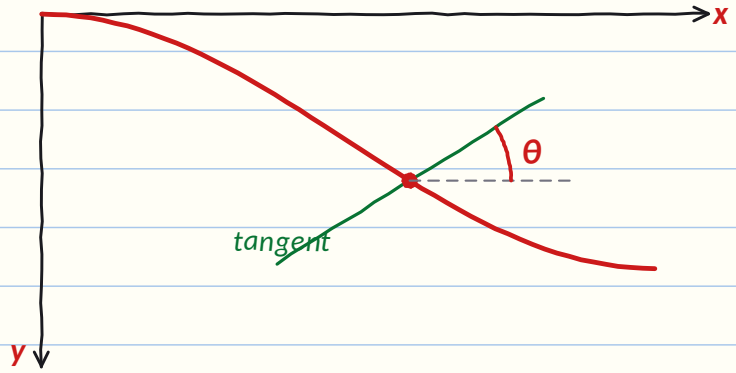
→ 4.1 Introduction to Deflection of Beams
→ 4.2 Double Integration Method – Cantilevers
→ 4.3 Double Integration – Simply Supported Beams
→ 4.4 Macaulay's Method
→ 4.5 Moment-Area Method (Mohr's Theorems)
→ 4.6 Standard Results and Problems

4.1 Introduction to Deflection of Beams

- Under load a beam bends – the axis takes a curved shape called the **elastic curve** (deflection curve).
- Deflection (y) : vertical displacement of a point on the axis. Slope (θ) : angle between the tangent to the elastic curve and the original axis.
- Why? Serviceability limits : $\delta_{max} \le L/325$ (typ. L/360 for floors, L/180 for cantilevers) ; excess deflection cracks plaster, damages machines.

Assumptions

- Beam is initially straight, material homogeneous & elastic (E const.), small deflections (slope is small so tan θ ≈ θ).
- Deflection due to shear is neglected ; plane sections remain plane.



Slope $\theta = dy/dx$

Curvature $1/R \approx d^2y/dx^2$

From bending theory $M / I = E / R$

$E\,I\,d^2y/dx^2 = M$

Integrate once → slope

Integrate twice → deflection

Differential equation of the elastic curve

- From geometry : $1/R = (d^2y/dx^2)/([1 + (dy/dx)^2]^{3/2}) \approx d^2y/dx^2$ (slope small).
- Bending equation : $M/I = E/R \Rightarrow 1/R = M/EI$.

$$E\,I\,\frac{d^2y}{dx^2} = M$$

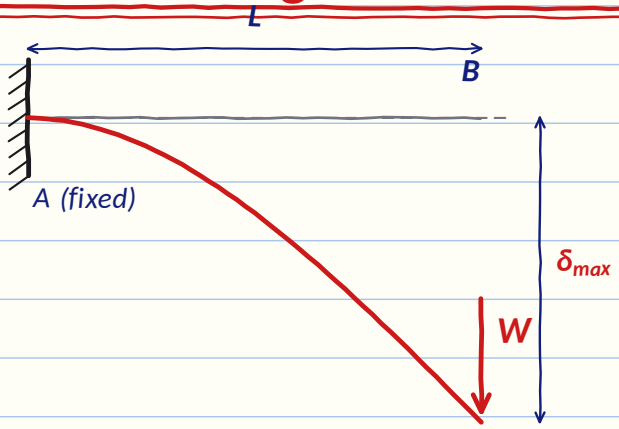
- Slope $\theta = dy/dx = 1/EI \int M\,dx + C_1$; deflection $y = 1/EI \iint M\,dx\,dx + C_1 x + C_2$.
- Sign : x → right, y ↑ +ve (so downward deflection is −ve), M sagging +ve. Constants C_1, C_2 from boundary conditions.

Support / point	Slope θ	Deflection y
Fixed end	0	0
Simple / roller support	≠ 0	0
Free end	≠ 0	≠ 0
Max deflection point	—	dy/dx = 0

Methods of finding slope and deflection

- (1) Double integration (2) Macaulay's method (3) Moment-area (Mohr's) method (4) Conjugate beam (5) Strain-energy (Castigliano).

4.2 Double Integration Method – Cantilevers



$M_x = -W(L - x)$ (x from fixed end A)

$EI\, d^2y/dx^2 = -W(L - x)$

slope & deflection at free end :

$\theta_B = W L^2 / 2EI$

$\delta_B = W L^3 / 3EI$

(a) Cantilever, point load W at free end B (fixed end A at x = 0)

- $M_x = -W(L - x) \Rightarrow EI\, y'' = -W(L - x)$
- $EI\, y' = -W(Lx - x^2/2) + C_1$; $y'(0) = 0 \Rightarrow C_1 = 0$
- $EI\, y = -W(Lx^2/2 - x^3/6) + C_2$; $y(0) = 0 \Rightarrow C_2 = 0$

$$\theta_B = \frac{W L^2}{2EI} \qquad \delta_B = \frac{W L^3}{3EI} \text{ (downward)}$$

(b) Cantilever with U.D.L. w over the whole span

- $M_x = -w(L - x)^2/2 \Rightarrow EI\, y'' = -w(L - x)^2/2$
- $EI\, y' = w(L - x)^3/6 + C_1$; $y'(0) = 0 \Rightarrow C_1 = -wL^3/6$
- $EI\, y = -w(L - x)^4/24 - (wL^3/6)x + C_2$; $y(0) = 0 \Rightarrow C_2 = wL^4/24$

$$\theta_B = \frac{w L^3}{6EI} \qquad \delta_B = \frac{w L^4}{8EI}$$

(c) Cantilever, load W at distance a from the fixed end

- Portion A–C bends ; portion C–B stays straight ($M = 0$) with slope θ_C.

$$\theta_C = \theta_B = \frac{W a^2}{2EI} \qquad \delta_C = \frac{W a^3}{3EI} \qquad \delta_B = \frac{W a^2}{6EI}(3L - a)$$

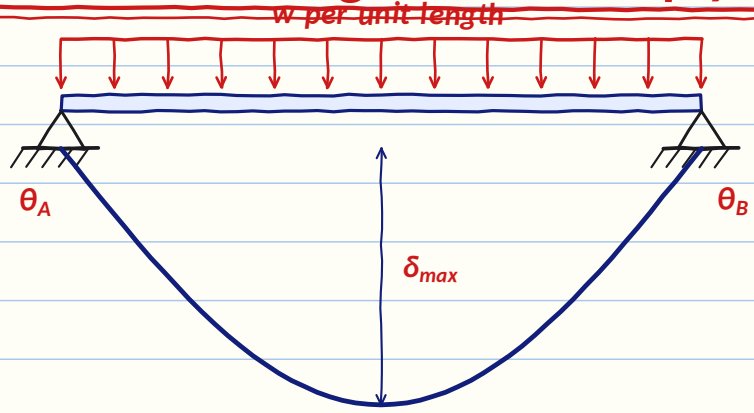
(d) Couple M at the free end

- $M_x = -M$ (constant) $\Rightarrow \theta_B = ML/EI$, $\delta_B = (ML^2)/2EI$.

(e) U.V.L. zero at free end, w at fixed end

- $M_x = - w (L - x)^3/6L \Rightarrow \theta_B = (w L^3)/24 E I$, $\delta_B = (w L^4)/30 E I$.

4.3 Double Integration – Simply Supported Beams



$M_x = w L x/2 - w x^2/2$

$\theta_A = -\theta_B = w L^3/24 E I$

$\delta_{max} = 5 w L^4/384 E I$ (at centre, $x = L/2$)

(a) Central point load W

- For $0 \le x \le L/2$: $M = W x/2 \Rightarrow E I y'' = W x/2$
- $E I y' = W x^2/4 + C_1$; slope = 0 at $x = L/2 \Rightarrow C_1 = - W L^2/16$
- $E I y = W x^3/12 - W L^2 x/16 + C_2$; $y(0) = 0 \Rightarrow C_2 = 0$

$$\theta_A = \theta_B = \frac{W L^2}{16 E I} \qquad \delta_{max} = \frac{W L^3}{48 E I} \text{ (at centre)}$$

(b) U.D.L. w over full span

- $M_x = w L x/2 - w x^2/2 \Rightarrow E I y' = w L x^2/4 - w x^3/6 + C_1$
- $E I y = w L x^3/12 - w x^4/24 + C_1 x + C_2$; $y(0)=0 \Rightarrow C_2 = 0$; $y(L)=0 \Rightarrow C_1 = - w L^3/24$

$$\theta_A = \theta_B = \frac{w L^3}{24 E I} \qquad \delta_{max} = \frac{5 w L^4}{384 E I}$$

(c) Eccentric point load W (a from A, b from B)

$$\delta_{under\ load} = \frac{W a^2 b^2}{3 E I L} \qquad \theta_A = \frac{W b (L^2 - b^2)}{6 L E I}$$

- Max deflection lies in the longer segment at $x = \sqrt{((L^2 - b^2)/3)}$ from A (for $a > b$).

(d) Couple M at end A (other end supported)

$$\theta_A = \frac{M L}{3 E I} \qquad \theta_B = \frac{M L}{6 E I} \qquad \delta_{max} = \frac{M L^2}{9\sqrt{3} E I} \text{ at } 0.577\ L$$

(e) U.V.L. (0 at A → w at B)

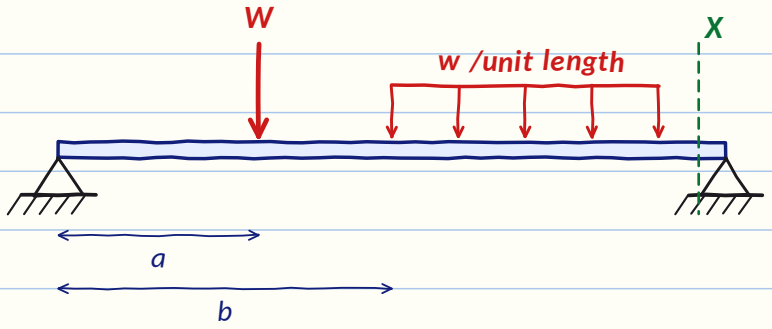
- $\theta_A = 7 w L^3/360 E I$; $\theta_B = w L^3/45 E I$; $\delta_{max} \approx 0.00652 w L^4/E I$ at 0.519 L from A.

Principle of superposition

When several loads act, slope/deflection = algebraic sum of the values due to each load separately (linear elastic, small deflection).

4.4 Macaulay's Method

Writes ONE bending-moment equation valid for the whole beam by using bracketed terms [x – a] that vanish when x < a. Integrate as a whole – constants C_1 and C_2 are common to all segments.



Take the section X–X in the LAST segment

Write M_x from the left end A

$M_x = R_A\, x - W\,[x-a] - w\,[x-b]^2/2$

Integrate everything as a whole;

[ ] term vanishes if x < its start

Rules

- Take the origin at the LEFT end ; write M_x for a section in the LAST segment (right-most).
- Point load W at a : – W [x – a] ; U.D.L. w beginning at b : – w $[x - b]^2/2$.
- A U.D.L. that stops before the end : extend it to the end and add an equal upward load + w $[x - c]^2/2$ from the stop point.
- Couple M_0 (clockwise) at d : + M_0 $[x - d]^0$.
- Integrate the brackets as $([x - a]^{n+1})/(n+1)$ – never expand them.
- A bracket whose value is negative (x < a) is dropped.

Example 1 : S.S. beam 6 m, W = 12 kN at 2 m from A (R_A = 8 kN)

$M_x = 8x - 12\,[x - 2]$

$E\,I\,y' = 4x^2 - 6\,[x - 2]^2 + C_1$

$E\,I\,y = (4/3)\,x^3 - 2\,[x - 2]^3 + C_1\,x + C_2$

$y(0) = 0 \Rightarrow C_2 = 0$; $y(6) = 0 \Rightarrow 288 - 128 + 6C_1 = 0 \Rightarrow$ **$C_1 = -26.67$**

Slope at A = – 26.67/EI ; deflection under load (x = 2) : E I y = 10.67 – 53.33 = **– 42.67 /EI kN·m³**

Max deflection where slope = 0 (x > 2) : $-2x^2 + 24x - 50.67 = 0 \Rightarrow x = 2.734$ m $\Rightarrow$ E I y = **– 46.45**

With E I = 10^4 kN·m² : $\delta_C = 4.27$ mm ; $\delta_{max} = 4.65$ mm ; $\theta_A = 0.00267$ rad

Example 2 : S.S. beam 6 m with U.D.L. 6 kN/m over the last 3 m (3 m ≤ x ≤

- Loads : total 18 kN at 4.5 m ⇒ R_B = 13.5 kN , R_A = 4.5 kN.

$M_x = 4.5\,x - 3\,[x-3]^2$

$E\,I\,y' = 2.25\,x^2 - [x-3]^3 + C_1$; $E\,I\,y = 0.375\,x^3 - 0.25\,[x-3]^4 + C_1\,x$

$y(6) = 0 \Rightarrow 81 - 20.25 + 6\,C_1 = 0 \Rightarrow$ **$C_1 = -10.125$**

$\theta_A = -10.125/EI$; $\theta_B = 2.25\times36 - 27 - 10.125 =$ **$+43.875/EI$**

Deflection at $x = 3$: $E\,I\,y = 10.125 - 30.375 =$ **$-20.25/EI$ kN·m^3**

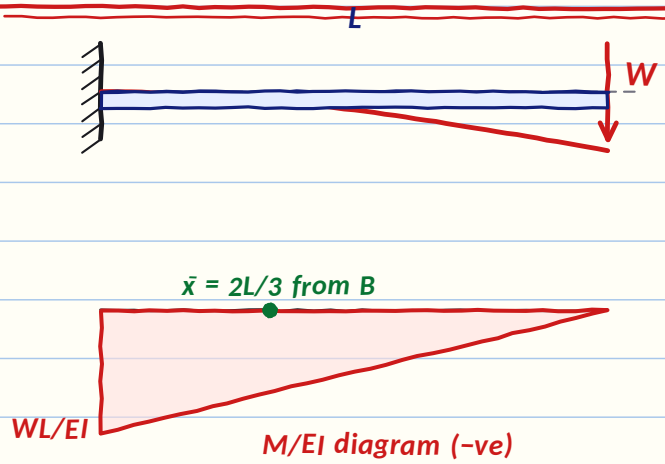
Overhanging beams

- Write M_x from the left end including both reactions ; boundary conditions : y = 0 at BOTH supports (not at the free end).
- Standard result – S.S. span L with overhang a carrying W at the free end :

$$\theta_B = \frac{W\,a\,L}{3\,E\,I} \qquad \theta_A = \frac{W\,a\,L}{6\,E\,I} \qquad y_C = \frac{W\,a^2(L+a)}{3\,E\,I} \quad \text{(overhang tip)}$$

- The portion between supports bends upward (hogging) – curvature reverses at the contra-flexure point.

4.5 Moment-Area Method (Mohr's Theorems)



Mohr's Theorem I :

$\theta_B - \theta_A$ = Area of M/EI diagram

Mohr's Theorem II :

$\delta = A\,\bar{x}$ (moment of area about B)

Here $A = \frac{1}{2} \cdot L \cdot WL/EI = W\,L^2/2EI$

$\delta_B = A \cdot \bar{x} = (W\,L^2/2EI)(2L/3) = W\,L^3/3EI$

Mohr's Theorem I : the change in slope between two points A and B of the elastic curve = area of the M/EI diagram between A and B.

$$\theta_B - \theta_A = \int_A^B \frac{M}{EI}\,dx = A_{M/EI}$$

Mohr's Theorem II : the deviation of point B from the tangent drawn at A = moment of the M/EI area between A and B about B.

$$\delta_{B/A} = A_{M/EI} \cdot \bar{x}_B$$

- $\bar{x}_B$ = distance of the centroid of the M/EI area from B. For a cantilever, tangent at the fixed end is horizontal ⇒ deviation = deflection.
- Useful areas : triangle ½ b h (centroid b/3 from the vertical side) ; parabola (vertex at the end) ⅓ b h (centroid 3b/4 from the vertex).

Example 3 : cantilever, W at free end

M/EI diagram – triangle, max = W L/EI at A, zero at B. Area = ½ × L × W L/EI = $W\,L^2/2EI$

θ_B = **$W\,L^2/2EI$** ; centroid 2L/3 from B ⇒ $\delta_B = (W\,L^2/2EI)(2L/3)$ = **$W\,L^3/3EI$**

Example 4 : cantilever with U.D.L.

$M/EI = w x^2/2EI$ (parabola, vertex at free end) ⇒ Area = ⅓ L ($w L^2/2EI$) = $w L^3/6EI$ = θ_B

centroid 3L/4 from B ⇒ $\delta_B = (w L^3/6EI)(3L/4) =$ **$w L^4/8EI$**

Example 5 : S.S. beam, central load

Tangent at C (midspan) is horizontal. θ_A = area of M/EI between A and C = ½ (L/2)(W L/4EI) = **$W L^2/16EI$**

δ_C = deviation of A from tangent at C = ($W L^2/16EI$)(L/3) = **$W L^3/48EI$**

4.6 Standard Results and Problems

Beam & loading	Max slope θ	Max deflection δ
Cantilever, W at free end	$W L^2/2EI$	$W L^3/3EI$
Cantilever, U.D.L. w	$w L^3/6EI$	$w L^4/8EI$
Cantilever, couple M at end	$M L/EI$	$M L^2/2EI$
Cantilever, U.V.L. (0 at tip)	$w L^3/24EI$	$w L^4/30EI$
S.S., W at centre	$W L^2/16EI$	$W L^3/48EI$
S.S., U.D.L. w	$w L^3/24EI$	$5 w L^4/384EI$
S.S., W at a, b	$W b (L^2-b^2)/6LEI$	$W a^2b^2/3EIL$
S.S., couple M at end	$M L/3EI$	$M L^2/9\sqrt{3}\ EI$

Example 6 (S.S. beam with U.D.L.)

$L = 4$ m, $w = 20$ kN/m, $E = 200$ GPa, $I = 1.2\times10^8$ mm^4.

$\delta_{max} = 5 w L^4/384 E I = 5\times20\times4000^4 / (384 \times 2\times10^5 \times 1.2\times10^8) =$ **2.78 mm**

Permissible $L/360 = 11.1$ mm ⇒ safe ✓ ; $\theta_A = w L^3/24EI =$

$20\times4000^3/(24\times2\times10^5\times1.2\times10^8) =$ **0.0022 rad**

Example 7 (Cantilever – superposition)

$L = 2$ m ; U.D.L. 5 kN/m over whole span + 8 kN at free end ; $E I = 4\times10^3$ kN·m^2.

$\delta_B = w L^4/8EI + W L^3/3EI = 80/32000 + 64/12000 = 0.0025 + 0.00533 =$ **7.83 mm**

Example 8 (Cantilever, moment-area)

$L = 2$ m, $W = 10$ kN at tip, $E I = 5\times10^3$ kN·m^2.

$\theta_B = W L^2/2EI = 10\times4/(2\times5000) =$ **0.004 rad** ; $\delta_B = W L^3/3EI = 80/15000 =$ **5.33 mm**

Example 9 (Cantilever, load not at the tip)

$L = 3$ m, $W = 6$ kN at $a = 1$ m from fixed end, $E I = 4\times10^3$ kN·m^2.

$\theta_C = \theta_B = W a^2/2EI = 6\times1/(2\times4000) =$ **0.00075 rad**

$\delta_C = W a^3/3EI = 6/12000 =$ **0.5 mm** ; $\delta_B = W a^2(3L - a)/6EI = 6\times8/24000 =$ **2.0 mm**

Example 10 (Macaulay with a couple)

S.S. beam 4 m, clockwise couple 8 kN·m at 1 m from A ⇒ $R_B = 2$ kN ↑, $R_A = 2$ kN ↓.

$M_x = -2x + 8\,[x - 1]^0$

$E I y' = -x^2 + 8\,[x - 1] + C_1$; $E I y = -x^3/3 + 4\,[x - 1]^2 + C_1 x$

$y(4) = 0 \Rightarrow -21.33 + 36 + 4 C_1 = 0 \Rightarrow$ **$C_1 = -3.667$**

$\theta_A = -3.667/EI$; deflection at the couple ($x = 1$) : $E I y = -0.333 - 3.667 =$ **-4.0**

/EI

Example 11 (Overhanging beam, standard result)

S.S. span $L = 4$ m, overhang $a = 1.5$ m, $W = 10$ kN at the tip, $EI = 2\times10^4$ kN·m^2.

$\theta_B = W a L/3EI = 10\times1.5\times4/(6\times10^4) =$ **0.001 rad**

$y_C = W a^2(L + a)/3EI = 10\times2.25\times5.5/(6\times10^4) =$ **2.06 mm** (downward at the tip)

Example 12 (Cantilever with U.D.L. – moment-area check)

$L = 2$ m, $w = 5$ kN/m, $EI = 4\times10^3$ kN·m^2.

$\theta_B = w L^3/6EI = 5\times8/24000 =$ **0.00167 rad** ; $\delta_B = w L^4/8EI = 80/32000 =$ **2.5 mm**

Steps to solve any deflection problem

- 1. Draw the free-body diagram ; find reactions.
- 2. Write M_x (Macaulay form if loads are not over the full span).
- 3. Integrate twice ; keep the constants C_1, C_2.
- 4. Apply boundary conditions (supports : $y = 0$; fixed end : $\theta = 0$).
- 5. Substitute x to get slope/deflection ; for y_{max} put $dy/dx = 0$.
- 6. Convert units : EI in kN·m^2 with lengths in m gives answer in m.

Short questions

- State Mohr's theorems. A: see 4.5 – slope change = M/EI area ; deviation = moment of M/EI area.
- What is the Macaulay bracket? A: $[x - a]^n = 0$ for $x < a$.
- Where is the deflection maximum? A: where slope $dy/dx = 0$.
- Boundary conditions for fixed end ? A: $\theta = 0$, $y = 0$.

STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – V

Columns, Thin and Thick Cylinders

Handwritten Study Notes

Syllabus topics covered in this unit :

Classification of columns — Euler's crippling load theory — end conditions — Equivalent length — Slenderness ratio — Limitations — Rankine-Gordon — Secant formula — Perry's formula — Thin & thick cylindrical shells — Hoop, longitudinal & volumetric strains — Lamé's theory — Compound cylinders

Contents of these notes :

→ 5.1 Columns – Introduction
→ 5.2 Euler's Crippling (Buckling) Load
→ 5.3 Rankine-Gordon Formula (Empirical)
→ 5.4 Eccentric Loading – Secant Formula
→ 5.5 Thin Cylindrical Shells
→ 5.6 Thick Cylinders – Lamé's Theory
→ 5.7 Compound Cylinders

5.1 Columns – Introduction

A column (strut) is a vertical compression member whose length is large compared with its lateral dimensions ; it fails by buckling (sudden lateral bending) before crushing.

Classification

- Short column : fails by crushing ; $\lambda < \sim 30$; $P = \sigma_c A$.
- Medium column : crushing + buckling ; Rankine/Johnson formulae.
- Long column : fails by buckling ; Euler's formula valid.
- Axially loaded (ideal) column vs eccentrically loaded column.

Slenderness ratio

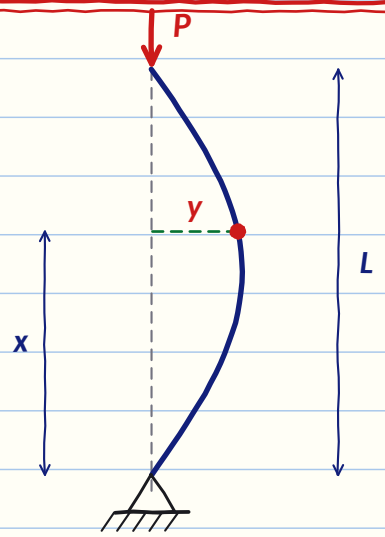
$$\lambda = \frac{L_e}{k} \qquad k = \sqrt{\left(\frac{I_{min}}{A}\right)} \quad \text{(least radius of gyration)}$$

- Column buckles about the axis of least I. L_e = equivalent (effective) length = length of the equivalent pin-ended column.

Assumptions of Euler's theory

- Column is perfectly straight, homogeneous, load strictly axial ; ends as specified.
- Material obeys Hooke's law ; stress ≤ proportional limit ; self-weight neglected ; failure only by buckling.
- Small deflections ; section uniform.

5.2 Euler's Crippling (Buckling) Load



At section x : $M = -P y$

$E I \, d^2y/dx^2 = -P y$

$d^2y/dx^2 + (P/EI)\, y = 0$

$y = A \cos \alpha x + B \sin \alpha x$, $\alpha^2 = P/EI$

$y(0) = 0 \Rightarrow A = 0$; $y(L) = 0 \Rightarrow \sin \alpha L = 0$

$\alpha L = n\pi \Rightarrow$ **$P_{cr} = \pi^2 E I / L^2$** $(n = 1)$

Derivation – both ends hinged

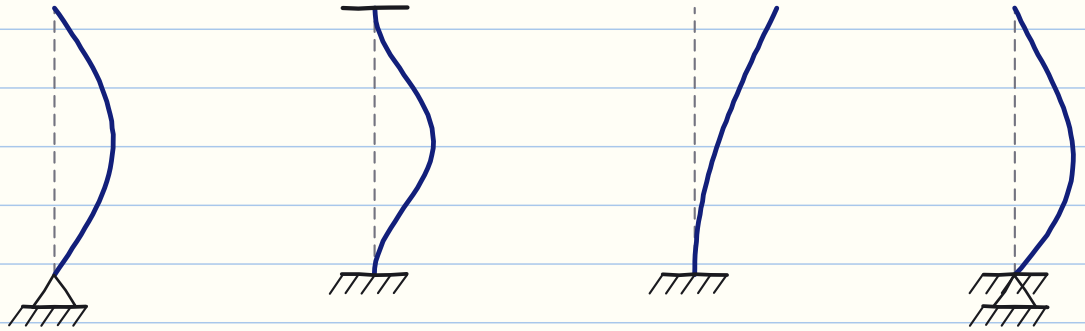
- Let y = deflection at distance x from the lower end ; moment $M = -P y$.
- $E I \, d^2y/dx^2 = -P y \Rightarrow d^2y/dx^2 + \alpha^2 y = 0$ with $\alpha^2 = P/EI$.
- Solution $y = A \cos \alpha x + B \sin \alpha x$. $y(0) = 0 \Rightarrow A = 0$; $y(L) = 0 \Rightarrow B \sin \alpha L = 0 \Rightarrow \alpha L$

$= n\pi$.

- Smallest load ($n = 1$) is the crippling load :

$$P_{cr} = \frac{\pi^2 E I}{L^2}$$

Other end conditions



Le = L	Le = L/2	Le = 2L	Le = L/√2
Both ends hinged	Both ends fixed	One fixed, one free	One fixed, one hinged

End conditions	L_e	Euler load P_{cr}
Both ends hinged	L	π^2EI / L^2
Both ends fixed	$L / 2$	$4\pi^2EI / L^2$
One fixed, other free	$2L$	$\pi^2EI / 4L^2$
One fixed, other hinged	$L/\sqrt{2}$	$2\pi^2EI / L^2$

$$P_{cr} = \frac{\pi^2 E I}{L_e^2}$$

- Fixed–free (derivation) : $EIy'' = P(\delta - y) \Rightarrow y = \delta(1 - \cos \alpha x)$; $y(L) = \delta \Rightarrow \cos \alpha L = 0 \Rightarrow \alpha L = \pi/2 \Rightarrow P = \pi^2EI/4L^2$.
- Ratio of strengths (hinged : fixed-fixed : fixed-free : fixed-hinged) = 1 : 4 : ¼ : 2.

Euler's critical stress

$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 E}{\lambda^2} \qquad \text{(hyperbola in } \sigma - \lambda \text{ plane)}$$

Limitations of Euler's formula

- Valid only while $\sigma_{cr} \le$ proportional limit $\Rightarrow \lambda \ge \lambda_c = \pi\sqrt{(E/\sigma_p)}$.
- For mild steel ($E = 2\times10^5$, $\sigma_p \approx 330$ N/mm^2) : $\lambda_c = \pi\sqrt{(2\times10^5/330)} = 77$; Euler applies for $\lambda > \sim 80$ (cast iron ~ 80, timber ~ 60).
- For small λ Euler gives absurdly high stress (> yield) ⇒ use Rankine/Johnson/Perry.
- Ignores initial crookedness, eccentricity, residual stress and imperfect end fixity.

Example 1 (Euler loads)

Solid circular column $d = 50$ mm, length 3 m, $E = 2\times10^5$ N/mm^2.

$I = \pi d^4/64 = 3.068\times10^5$ mm^4 ; $k = d/4 = 12.5 \Rightarrow \lambda = 3000/12.5 = 240$ (long ✓)

Both ends hinged : $P_{cr} = \pi^2 \times 2\times10^5 \times 3.068\times10^5 / 3000^2 =$ **67.3 kN**

Both ends fixed : $4 \times 67.3 =$ **269.2 kN** ; fixed–free : $67.3/4 =$ **16.8 kN** ;

fixed–hinged : $2 \times 67.3 =$ **134.6 kN**

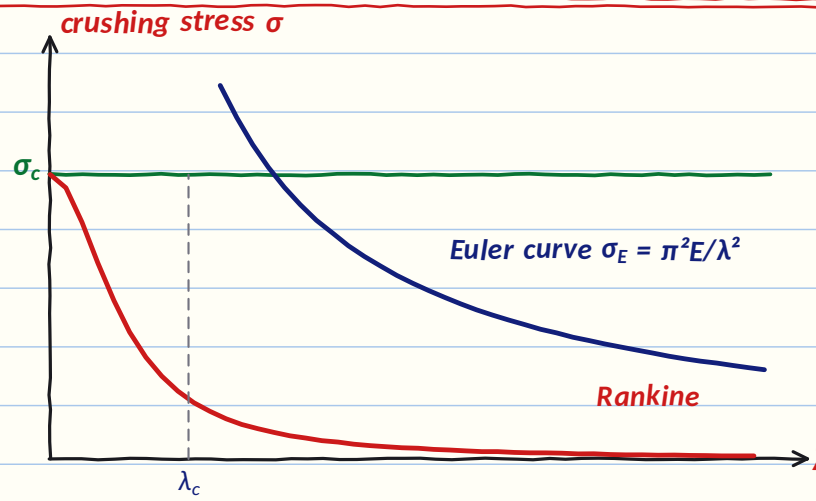
Example 2 (Hollow / rectangular section)

Rectangular column 60 × 100 mm, 2.5 m, both ends hinged, $E = 2\times10^5$:

$I_{min} = 100 \times 60^3/12 = 1.8\times10^6$ mm^4 (buckles about the weaker axis)

$P_{cr} = \pi^2 \times 2\times10^5 \times 1.8\times10^6 / 2500^2 =$ **568.4 kN**

5.3 Rankine–Gordon Formula (Empirical)



Rankine–Gordon :

$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_E}$

$P_R = \sigma_c A /(1 + a\,\lambda^2)$

$a = \sigma_c /\pi^2 E$ (Rankine const.)

Short: λ small → $P \approx$ crushing

Long: λ large → $P \approx$ Euler

- Covers short, medium and long columns. Rankine's crippling load P_R :

$$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_E} \qquad P_c = \sigma_c A \ \text{(crushing)}, \ P_E = \text{Euler load}$$

$$P_R = \frac{\sigma_c A}{1 + a\,\lambda^2} \qquad a = \frac{\sigma_c}{\pi^2 E}$$

Material	σ_c (N/mm^2)	a
Mild steel	330	1/7500
Cast iron	550	1/1600
Wrought iron	250	1/9000
Timber	50	1/750

- λ small ⇒ $P_R \to \sigma_c A$ (crushing) ; λ large ⇒ $P_R \to$ Euler load.

Example 3 (Rankine)

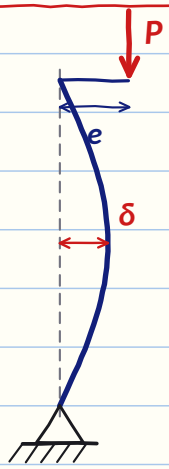
Solid circular steel column $d = 100$ mm, 3 m, both ends hinged, $\sigma_c = 330$ N/mm^2, $a = 1/7500$.

$A = 7854$ mm^2 ; $k = 25$ mm ; $\lambda = 3000/25 = 120$

$P_R = 330 \times 7854 / (1 + 120^2/7500) = 2.592\times10^6 / 2.92 =$ **887.6 kN**

Safe load (F.O.S. 3) = **295.9 kN**

5.4 Eccentric Loading – Secant Formula



Max deflection $\delta = e\,[\,\sec(\tfrac{1}{2}\sqrt{(P\,L^2/EI)}) - 1\,]$

Max moment $M_{max} = P\,(e + \delta)$

$$\sigma_{max} = \frac{P}{A}\left(1 + \frac{e\,c}{k^2}\sec\left(\frac{L}{2k}\sqrt{\frac{P}{AE}}\right)\right)$$

SECANT FORMULA (pin-ended, $c = y_{max}$)

- Load P acts at eccentricity e from the axis of a pin-ended column ; bending moment at mid-height = $P\,(e + \delta)$.

$$\delta = e\left[\sec\left(\frac{L}{2}\sqrt{\frac{P}{EI}}\right) - 1\right]$$

$$\sigma_{max} = \frac{P}{A}\left[1 + \frac{e\,c}{k^2}\sec\left(\frac{L_e}{2k}\sqrt{\frac{P}{A\,E}}\right)\right]$$

- c = distance of extreme fibre from the axis ; $e\,c/k^2$ = eccentricity ratio. Failure when $\sigma_{max} = \sigma_y$.
- Because P appears inside sec, the equation is solved by trial for P.

Example 4 (Secant)

Pin-ended column 100 × 100 mm, 3 m, $P = 100$ kN, $e = 10$ mm, $E = 2\times10^5$.

$k^2 = 100^2/12 = 833.3$; $c = 50$; $e\,c/k^2 = 0.6$

$(L/2k)\sqrt{(P/AE)} = (3000/57.74) \times \sqrt{(10^5/(10^4\times2\times10^5))} = 0.3674$ rad $\Rightarrow \sec = 1.0715$

$\sigma_{max} = 10\,[1 + 0.6 \times 1.0715] =$ **16.4 N/mm^2**

Perry's formula (Perry–Robertson)

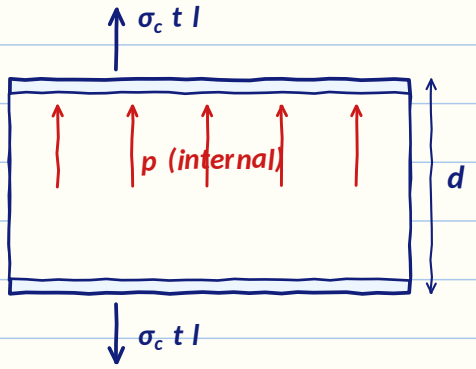
$$(\sigma_y - \sigma_a)(\sigma_E - \sigma_a) = \eta\, \sigma_E\, \sigma_a$$

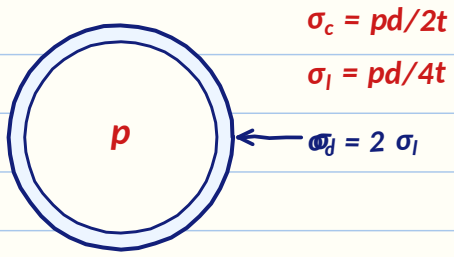
- σ_a = average axial stress at failure = P/A ; σ_y = yield stress ; σ_E = Euler stress = $\pi^2 E/\lambda^2$.
- η = Perry factor = $0.3\,(\lambda/100)^2$ (allows for initial curvature / eccentricity) – basis of many design codes.
- Solution : $\sigma_a = \tfrac{1}{2}[\,\sigma_y + (1+\eta)\,\sigma_E - \sqrt{((\sigma_y + (1+\eta)\sigma_E)^2 - 4\,\sigma_y\,\sigma_E)}\,]$.

5.5 Thin Cylindrical Shells

Thin shell : wall thickness t < d/20 ; stress assumed uniform across thickness.

Internal pressure p.





Half cylinder: $2\,\sigma_c\,t\,l = p\,d\,l$ End: $\sigma_l\,\pi\,d\,t = p\,\pi\,d^2/4$

Circumferential (hoop) stress

- Consider a length l ; force from pressure on half-cylinder = $p\,d\,l$; resisted by $2\,\sigma_c\,t\,l$.

$$\sigma_c = \frac{p\,d}{2\,t}$$

Longitudinal stress

- End force $p\,\pi\,d^2/4$ resisted by wall area $\pi\,d\,t$:

$$\sigma_l = \frac{p\,d}{4\,t} = \frac{\sigma_c}{2}$$

- Max shear stress (in plane) = $(\sigma_c - \sigma_l)/2 = p\,d/8t$.

Strains and volume change (μ = Poisson's ratio)

$$\varepsilon_c = \frac{p\,d}{4\,t\,E}(2-\mu) \qquad \varepsilon_l = \frac{p\,d}{4\,t\,E}(1-2\mu)$$

$$\varepsilon_v = 2\,\varepsilon_c + \varepsilon_l = \frac{p\,d}{4\,t\,E}(5-4\mu)$$

- Change in diameter $\delta d = \varepsilon_c\,d$; change in length $\delta L = \varepsilon_l\,L$; $\delta V = \varepsilon_v\,V$.
- Joint efficiency : $\sigma_c = p\,d\,/\,(2\,t\,\eta_l)$ (longitudinal joint) ; $\sigma_l = p\,d\,/\,(4\,t\,\eta_c)$ (circumferential joint).
- Thin spherical shell : $\sigma = p\,d/4t$; $\varepsilon = p\,d/4tE(1-\mu)$; $\varepsilon_v = 3\varepsilon = 3\,p\,d/4tE(1-\mu)$.

Example 5

Thin cylinder $d = 1.2$ m, $t = 12$ mm, $p = 1.5$ N/mm^2, $L = 3$ m, $E = 2\times10^5$, $\mu = 0.3$.

$\sigma_c = 1.5 \times 1200/(2\times12) =$ **75 N/mm^2** ; $\sigma_l =$ **37.5 N/mm^2**

$\varepsilon_c = (75 - 0.3\times37.5)/2\times10^5 = 3.19\times10^{-4}$; $\varepsilon_l = (37.5 - 0.3\times75)/2\times10^5 = 7.5\times10^{-5}$

$\delta d = 3.19\times10^{-4} \times 1200 =$ **0.383 mm** ; $\delta L = 7.5\times10^{-5} \times 3000 =$ **0.225 mm**

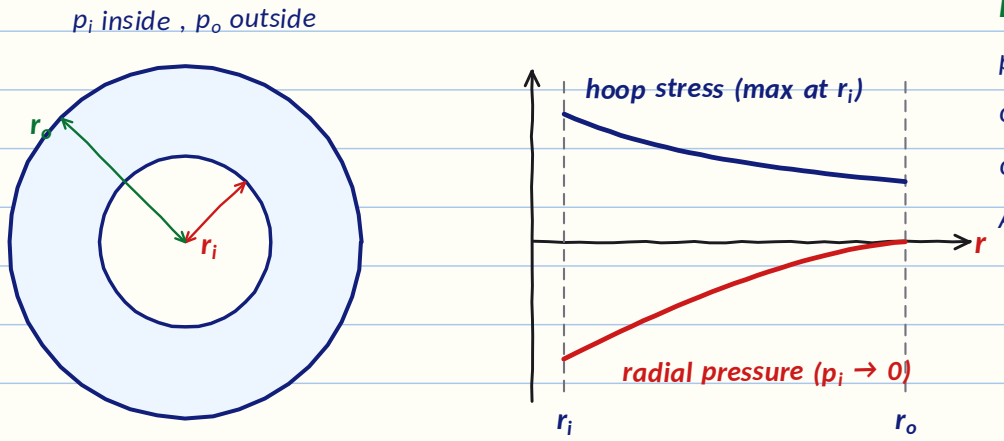
$\varepsilon_v = 2(3.19\times10^{-4}) + 7.5\times10^{-5} = 7.13\times10^{-4}$; $V = 3.393\times10^9$ mm^3 $\Rightarrow \delta V =$ **2.42×10^6 mm^3**

Example 6 (Thickness)

$d = 800$ mm, $p = 2.5$ N/mm^2, $\sigma_{perm} = 100$ N/mm^2 : $t = p\,d/2\sigma = 2.5\times800/200 =$ **10 mm**

5.6 Thick Cylinders – Lamé's Theory

Thick shell : $t > d/20$; hoop stress σ_h varies across the thickness and radial stress p_r acts.



Lamé :

$p_r = A/r^2 - B$

$\sigma_h = A/r^2 + B$

$\sigma_h + p_r = 2A/r^2$

A, B from B.C.

Assumptions

- Material homogeneous, isotropic, elastic ; plane sections remain plane (longitudinal strain constant) ; σ_l uniform.

Derivation of Lamé's equations

- Element between r and r + dr (angle dθ). Radial balance (p_r compressive +) : $p_r\, r\, d\theta = (p_r + dp_r)(r + dr)d\theta + \sigma_h\, dr\, d\theta$
- $\Rightarrow \sigma_h = -\left(p_r + r\, dp_r/dr \right)$ … (1)
- Constant longitudinal strain $\Rightarrow \sigma_h - p_r = 2B$ (constant) … (2). Put (2) in (1) : $r\, dp_r/dr = -2(p_r + B) \Rightarrow p_r + B = A/r^2$. Hence :

$$p_r = \frac{A}{r^2} - B \qquad \sigma_h = \frac{A}{r^2} + B$$

- Boundary conditions : $p_r = p_i$ at $r = r_i$; $p_r = p_o$ at $r = r_o$.

$$A = \frac{(p_i - p_o)\, r_i^2\, r_o^2}{r_o^2 - r_i^2} \qquad B = \frac{p_i\, r_i^2 - p_o\, r_o^2}{r_o^2 - r_i^2}$$

Only internal pressure ($p_o = 0$)

$$\sigma_h = \frac{p_i\, r_i^2}{r_o^2 - r_i^2}\left(1 + \frac{r_o^2}{r^2}\right) \qquad p_r = \frac{p_i\, r_i^2}{r_o^2 - r_i^2}\left(\frac{r_o^2}{r^2} - 1\right)$$

- Max hoop stress at the inner surface : $\sigma_{h,max} = p_i\,(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$ ($> p_i$ always).
- At outer surface : $\sigma_h = 2\, p_i\, r_i^2/(r_o^2 - r_i^2)$; radial pressure falls from p_i to 0.
- Longitudinal stress (closed ends) $\sigma_l = p_i\, r_i^2/(r_o^2 - r_i^2) = B$.
- Design thickness for allowable σ_t : $r_o = r_i \sqrt{((\sigma_t + p)/(\sigma_t - p))}$.

Example 7

r_i = 100 mm, r_o = 150 mm, p_i = 80 N/mm^2, p_o = 0.

$B = 80 \times 10^4/(22500 - 10000) = 64$; $A = B\, r_o^2 = 1.44 \times 10^6$

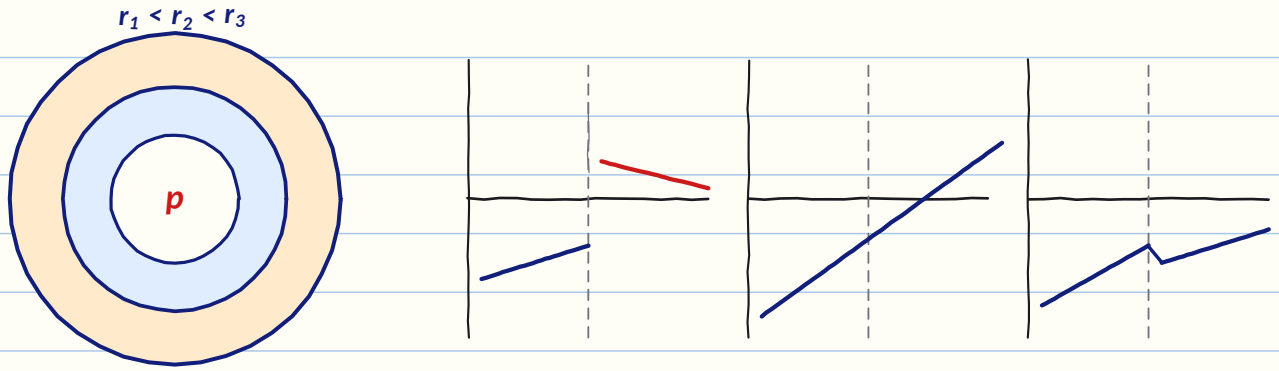
σ_h (inner) = $A/r_i^2 + B = 144 + 64 =$ **208 N/mm^2** ; σ_h (outer) = 64 + 64 = **128 N/mm^2**

p_r : 80 N/mm^2 (inner) → 0 (outer) ; σ_l = **64 N/mm^2**

Example 8 (Wall thickness)

Inner radius 80 mm, p = 40 N/mm^2, σ_t = 120 N/mm^2 : $r_o = 80\sqrt{(160/80)} = 113.1$ mm ⇒ **t = 33.1 mm**

5.7 Compound Cylinders



inner tube + outer jacket (shrunk on) hoop stress variation across the wall (schematic)

- Outer jacket (smaller bore) is heated and shrunk on the inner tube ⇒ inner tube is compressed, outer jacket is in hoop tension *before* loading.
- Resultant stress = shrinkage stress + stress due to working pressure (Lamé) ⇒ stress is far more uniform across the wall and material is better used.
- Junction pressure (shrinkage) p_s acts at r_2 ; radii r_1 (bore), r_2 (junction), r_3 (outside).
- Inner tube : σ_h at $r_2 = -\,p_s\,(r_2^2 + r_1^2)/(r_2^2 - r_1^2)$; at $r_1 = -\,2\,p_s\,r_2^2/(r_2^2 - r_1^2)$.
- Outer tube : σ_h at $r_2 = +\,p_s\,(r_3^2 + r_2^2)/(r_3^2 - r_2^2)$; at $r_3 = +\,2\,p_s\,r_2^2/(r_3^2 - r_2^2)$.
- Radial interference (same material) $\approx r_2\,(\sigma_{h,\text{outer-bore}} + |\sigma_{h,\text{inner-outside}}|)/E$.

Procedure

- 1. Draw Lamé distribution for the shrinkage pressure p_s (inner tube compressive, outer tube tensile).
- 2. Draw the Lamé distribution for the internal working pressure on the equivalent single cylinder.
- 3. Algebraically add the two ⇒ resultant hoop stress at each radius.
- 4. Choose p_s so that max resultant hoop stress is equal in both tubes (best design).

Example 9

$r_1 = 100$, $r_2 = 150$, $r_3 = 200$ mm ; $p_s = 20$ N/mm^2, $E = 2\times10^5$.

Outer tube : bore $20(40000 + 22500)/(17500) =$ **71.4 N/mm^2** (tension) ; outside $2\times20\times22500/17500 =$ **51.4 N/mm^2**

Inner tube : outside $-\,20(32500)/12500 =$ **– 52 N/mm^2** ; bore $-\,2\times20\times22500/12500 =$ **– 72 N/mm^2** (compression)

Interference $\approx 150\,(71.4 + 52)/2\times10^5 =$ **0.093 mm** on radius

Formula summary – Unit V

Item	Formula
Euler load	$\pi^2 EI / L_e^2$
Slenderness ratio	$\lambda = L_e / k$
Rankine	$\sigma_c A /(1 + a \lambda^2)$
Thin cylinder hoop	$\sigma_c = p\ d / 2t$
Thin cylinder long.	$\sigma_l = p\ d / 4t$
Lamé hoop stress	$\sigma_h = A/r^2 + B$
Lamé radial pressure	$p_r = A/r^2 - B$

Short questions

- Define slenderness ratio. A: effective length / least radius of gyration.
- Why does a column buckle about the weaker axis? A: least I gives least P_{cr}.
- Hoop stress vs longitudinal stress in thin cylinder? A: $\sigma_c = 2\ \sigma_l$.
- Where is the hoop stress maximum in a thick cylinder? A: at the inner surface.